MULTI-ANGLE DYNAMIC LIGHT SCATTERING EXPERIMENTS WITH REAL-TIME AUTOCORRELATION USING FIELD-PROGRAMMABLE GATE ARRAYS

A Thesis

Presented in

Partial Fulfillment of the

Requirements for the Degree of

MASTER OF SCIENCE

August, 2 0 2 6

BY

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ACKNOWLEDGEMENTS

I would like to first thank Dr. Landahl for his guidance throughout this project. I learned how difficult it was to build an experiment from the ground up, and I came to appreciate that process. Thank you for the opportunity to learn and to teach others.

For their collaboration and help in creating a functioning chemistry station, and for allowing me to become a mentor, I would like to thank Isabelle Anderson, Ky Blake, Alejandro Campos, and Lucas Rodriguez.

I would like to thank the DePaul Department of Physics for providing a supportive and engaging environment. In particular, I thank Dr. Anuj Sarma and Dr. Bernhard Beck-Winchatz for serving on my thesis committee, and Gabriel Williford for developing the FPGA correlation device that made this work possible.

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ABSTRACT

Dynamic light scattering (DLS) determines particle size from temporal fluctuations in coherently scattered light. Here a field-programmable gate array (FPGA) computes intensity autocorrelations in real time, and an automated diffractometer—with custom 3D-printed cuvette, laser, homing, and GT2-gear mounts—enables multi-angle measurements. Square-wave timing tests at 1 Hz, 100 Hz, and 10 kHz verified the correlator response. Calibrated latex spheres were then measured at seven scattering angles from 32.5° to 44.5°. Each autocorrelation function was fit to the standard single-exponential form $g(\tau) = B + \beta e^{-2\Gamma\tau}$ with weights from analytic per-lag Poisson uncertainties $\sigma_g = \sqrt{g/N}$ based on coincidence counts. The extracted decay rates Γ varied linearly with q^2, yielding diffusion coefficients and hydrodynamic diameters of 94 ± 8, 397 ± 28, and 556 ± 64 nm for latex spheres of nominal size 90, 300, and 490 nm. Motor-induced noise was identified and removed by disabling the stepper drive during acquisition. These results support FPGA-based real-time correlation as a viable approach for quantitative multi-angle DLS at the proof-of-concept level.

CHAPTER 1

Introduction

1.1 Thesis motivation

Dynamic light scattering (DLS) determines particle size by measuring temporal fluctuations in coherent scattered light caused by Brownian motion. Conventional DLS instruments determine the particle diffusion coefficient from the decay of the intensity autocorrelation function measured at one or more scattering angles. This thesis investigates whether a real-time field-programmable gate array (FPGA) autocorrelator can perform quantitative multi-angle DLS measurements. Validation is achieved by demonstrating that the measured decay rate varies linearly with the squared magnitude of the scattering vector (Fig. 1.5), allowing a single diffusion coefficient and hydrodynamic diameter to be recovered from measurements at several detector angles.

The increasing temporal resolution of photon-correlation measurements places greater demands on both data acquisition and analysis. The development of higher-brilliance synchrotron and free-electron-laser sources is expected to extend X-ray photon correlation spectroscopy (XPCS) to sub-microsecond timescales and substantially broaden the range of dynamics that can be measured [1]. High-repetition-rate X-ray sources require detection systems capable of resolving closely spaced photon pulses, with detector dead-time becoming an important consideration at count rates approaching hundreds of MHz [2]. At the same time, the large volume of data generated by MHz-rate measurements presents significant challenges for correlation analysis, requiring specialized data-processing and analysis methods

to efficiently extract dynamic information [3]. These developments demonstrate the need for high-speed processing techniques that can perform correlation calculations efficiently as measurement rates increase, motivating the use of hardware-based approaches such as field-programmable gate arrays (FPGAs).

A field-programmable gate array (FPGA) is a programmable digital device that implements custom hardware logic rather than executing instructions sequentially as a conventional processor. Because many logic operations occur simultaneously, FPGAs are well suited for high-speed real-time signal processing applications such as photon correlation.

Williford demonstrated that an FPGA can compute intensity autocorrelation functions in real time by continuously processing photon arrival events as they are detected [4]. Rather than storing complete photon time histories for later analysis, the FPGA performs the correlation calculation directly in hardware, providing immediate access to the autocorrelation function (Fig. 1.1).

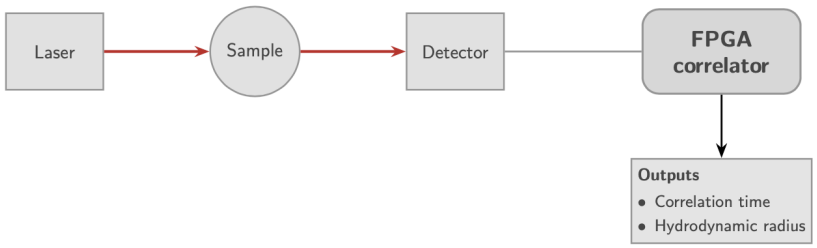


Figure 1.1: An illustration of a dynamic light scattering (DLS) test bench. Scattered light from the sample is collected by a detector and analyzed by an FPGA-based autocorrelator. The autocorrelation function is then used to determine the characteristic correlation time and calculate the hydrodynamic radius of the particles.

Previous work demonstrated that the FPGA autocorrelator could recover reasonable particle sizes from fixed-angle DLS measurements. However, fixed-angle measurements alone cannot distinguish correct physical behavior from systematic errors such as imperfect alignment, detector artifacts, concentration effects, or fitting errors. A

more stringent validation is obtained by measuring the autocorrelation function at multiple scattering angles. Although the decay rate changes with scattering vector, all measurements should yield the same diffusion coefficient. Demonstrating this relationship is the principal objective of this thesis.

This thesis addresses the challenges of creating a multi-angle test bench for quantitative dynamic light scattering. Custom 3D-printed mounts were used to align and center the motorized diffractometer and detector around the sample so that the detector observes the same scattering volume throughout data collection. The remainder of this chapter reviews the physical basis of DLS, the motorized diffractometer, and the custom printed parts that realize its geometry (with a full catalog in Appendix A). Chapter 2 describes sample preparation and operating procedures. Chapter 3 presents the square-wave, latex, motor-noise, and coffee-creamer measurements. Chapter 4 interprets those results and the uncertainty budget, and Chapter 5 states the overall conclusion.

1.2 Brownian motion

Brownian motion arises because suspended particles undergo continual collisions with the surrounding molecules of the fluid. These collisions produce a random walk whose mean-squared displacement increases linearly with time. In DLS, this random motion changes the relative phases of the light scattered by different particles, causing the observed speckle pattern to fluctuate in time.

The fluctuating speckle pattern is described by the decay rate Γ. The decay rate is related to the particle diffusion coefficient D and the scattering vector magnitude q through

$$\Gamma = Dq^2, \tag{1.1}$$

where D is the diffusion coefficient. The magnitude of the scattering vector is

$$q = |\mathbf{q}| = \frac{4\pi n}{\lambda} \sin\left(\frac{\theta}{2}\right). \tag{1.2}$$

In Eq. 1.2, λ is the wavelength of the incident light and θ is the detector angle. The refractive index of the liquid, n, accounts for light traveling through the liquid. The detector angle is the difference between the transmitted laser beam and the path that scattered light takes to the detector (Fig. 1.2).

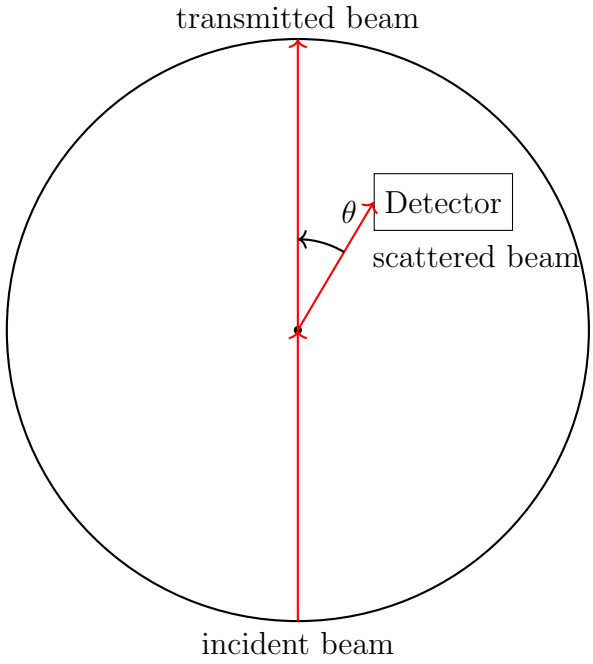


Figure 1.2: Scattering geometry for multi-angle DLS. The incident beam enters the sample from below; the transmitted beam continues along the same axis. The detector collects scattered light at angle θ relative to the transmitted beam while remaining on a circular path centered on the sample.

At larger scattering angles, a given particle displacement produces a larger phase change between scattered waves, causing the speckle pattern to decorrelate more rapidly. Consequently, the decay rate increases as the scattering vector increases. This angular dependence provides the main experimental test of whether the measured fluctuations are caused by Brownian diffusion.

1.3 Scattering of a coherent source

Scattering of coherent light from small particles provides the optical signal for DLS. The latex standards used in this experiment are polystyrene latex microspheres containing a single particle size and uniformity described by the manufacturer(Tab. 2.2). The smallest latex standards used here ($r \ll \lambda$) are in the Rayleigh regime of dipole radiation, while the larger standards approach the Mie regime; in either case the sizing result depends on the temporal decay of the intensity correlation, not on the absolute scattering cross section. As a plane wave encounters a particle, it is re-radiated as a spherical wave. Figure 1.3 shows this geometry for a plane wave traveling in the z direction and a single particle.

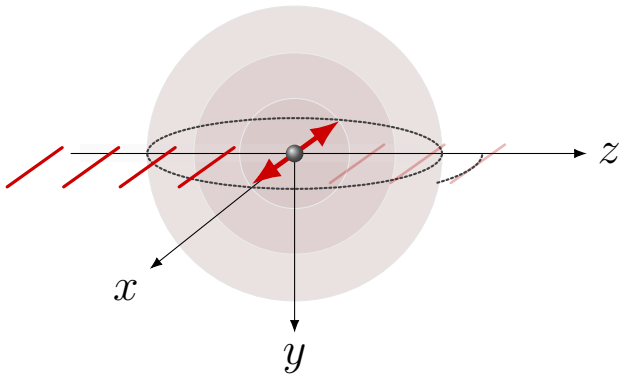


Figure 1.3: From the left, a red plane wave propagates along the z-axis. Upon encountering a particle, the particle emits a spherical wave of the same wavelength.

Although the scattered light spreads in three dimensions, it is common to work in a two-dimensional plane. With the origin chosen so that $y = 0$, the scattering-vector magnitude q is given by Eq. 1.2, where θ is the angle between the transmitted beam and the detector.

Each nanoparticle acts like a scatterer, converting incident light of any polarization into outgoing spherical waves. Scattered light from different nanoparticles can interfere constructively or destructively. In DLS, that interference arises from relative

phase differences between scattered waves. These temporal fluctuations in intensity caused by particles acting as scatterers are measured.

The experimental geometry is shown in Fig. 1.4. The incident wave propagates with wavevector $\boldsymbol{k}_i$, and light collected by a detector propagates from the scattering volume with wavevector $\boldsymbol{k}_s$. The scattering angle θ is the angle between the incident and detected propagation directions. The detector therefore selects the scattering angle and, through that angle, the magnitude of the scattering vector. The scattering vector itself is defined in reciprocal space and is not a real-space distance in the apparatus.

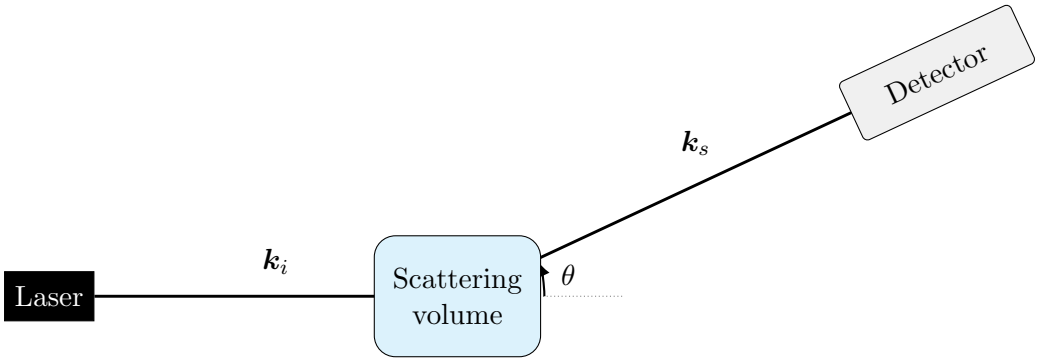


Figure 1.4: Real-space scattering geometry. The detector position selects the scattered propagation direction $\boldsymbol{k}_s$ and hence the scattering angle θ relative to the incident direction $\boldsymbol{k}_i$. The angle θ is a real-space angle, but $\boldsymbol{q}$ is not a real-space distance.

The scattering vector is defined in reciprocal space rather than as a distance between components of the apparatus. Using the convention adopted here and in the thesis,

$$\boldsymbol{q} = \boldsymbol{k}_s - \boldsymbol{k}_i. \tag{1.3}$$

The corresponding vector construction is shown in Fig. 1.5. For elastic scattering, $|\boldsymbol{k}_i| = |\boldsymbol{k}_s| = k$, where $k = 2\pi n/\lambda_0$. The three vectors therefore form an isosceles triangle. If the angle θ is bisected, the resulting right triangle has hypotenuse k, angle $\theta/2$, and opposite side $q/2$. It follows directly that

$$\sin\left(\frac{\theta}{2}\right) = \frac{q/2}{k}, \tag{1.4}$$

and hence

$$q \equiv |\boldsymbol{q}| = 2k \sin\left(\frac{\theta}{2}\right) = \frac{4\pi n}{\lambda_0} \sin\left(\frac{\theta}{2}\right), \tag{1.5}$$

where n is the refractive index of the sample medium and λ_0 is the wavelength in vacuum.

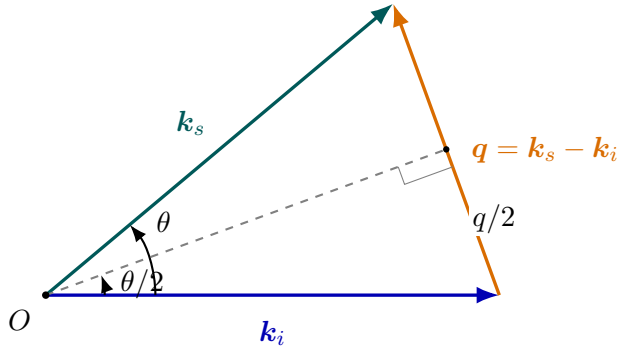


Figure 1.5: Reciprocal-space construction of the scattering vector using the thesis convention $\boldsymbol{q} = \boldsymbol{k}_s - \boldsymbol{k}_i$. Accordingly, $\boldsymbol{q}$ points from the tip of $\boldsymbol{k}_i$ to the tip of $\boldsymbol{k}_s$. The dashed bisector produces a right triangle containing the angle $\theta/2$ and the side $q/2$, which gives $q = 2k \sin(\theta/2)$. The arrow lengths represent wavevector magnitudes, not physical path lengths or detector positions.

For particles undergoing translational Brownian diffusion, let $\rho(\boldsymbol{r}, t)$ denote the local particle-number density. This density field obeys Fick's second law,

$$\frac{\partial \rho(\boldsymbol{r}, t)}{\partial t} = D \nabla^2 \rho(\boldsymbol{r}, t), \tag{1.6}$$

where D is the translational diffusion coefficient. Define the spatial Fourier component of the density by

$$\rho(\boldsymbol{q}, t) \equiv \int \rho(\boldsymbol{r}, t) \exp(-i\boldsymbol{q}\cdot\boldsymbol{r})\, d^3 r. \tag{1.7}$$

Taking the Fourier transform of Eq. (1.6), and assuming that the boundary terms vanish, gives

$$\frac{\partial \rho(\boldsymbol{q}, t)}{\partial t} = D \int \nabla^2 \rho(\boldsymbol{r}, t) \exp(-i\boldsymbol{q}\cdot\boldsymbol{r})\, d^3 r = -Dq^2 \rho(\boldsymbol{q}, t). \tag{1.8}$$

Thus, during a delay time τ, each density mode evolves according to

$$\rho(\boldsymbol{q}, t + \tau) = \rho(\boldsymbol{q}, t) \exp(-Dq^2 \tau). \tag{1.9}$$

The density-density correlation at wavevector $\boldsymbol{q}$ is the intermediate scattering function

$$F(q,\tau) \equiv \langle \rho(-\boldsymbol{q},t)\rho(\boldsymbol{q},t+\tau)\rangle\,, \tag{1.10}$$

where the brackets denote an ensemble average. Substitution of Eq. (1.9) yields

$$F(q,\tau) = F(q,0)\exp(-Dq^2\tau). \tag{1.11}$$

The normalized intermediate scattering function is therefore

$$f(q,\tau) \equiv \frac{F(q,\tau)}{F(q,0)} = \exp(-Dq^2\tau). \tag{1.12}$$

In the single-scattering approximation, the scattered electric-field amplitude at $\boldsymbol{q}$ is proportional to the corresponding density Fourier component: $E_s(\boldsymbol{q},t) \propto \rho(\boldsymbol{q},t)$. The normalized electric-field autocorrelation function is consequently

$$g^{(1)}(q,\tau) \equiv \frac{\langle E_s^*(\boldsymbol{q},t)E_s(\boldsymbol{q},t+\tau)\rangle}{\langle |E_s(\boldsymbol{q},t)|^2\rangle} \tag{1.13}$$

$$= \frac{F(q,\tau)}{F(q,0)} = \exp(-Dq^2\tau) = \exp(-\Gamma\tau), \tag{1.14}$$

so that

$$\Gamma = Dq^2. \tag{1.1}$$

Thus, at a fixed scattering angle and wavelength, the field-correlation decay rate is directly proportional to the diffusion coefficient. Through the Siegert relation, the measured normalized intensity correlation function is

$$g^{(2)}(q,\tau) = 1 + \beta\left|g^{(1)}(q,\tau)\right|^2 = 1 + \beta\exp(-2\Gamma\tau), \tag{1.15}$$

where β is the coherence factor of the instrument. Equation (1.15) makes clear that $\Gamma = Dq^2$ is the decay rate of the field correlation function; the exponential in $g^{(2)}-1$ contains 2Γ. This distinction should be retained when reporting a fitted decay rate.

1.4 Autocorrelation

The measured intensity correlation function in a dynamic light scattering experiment is $G_2(\tau)$. This function links microscopic molecular motions to fluctuations in scattered light intensity over time [5]. It is a second-order correlation function given by the time-averaged product of the intensity $I(t)$ and the intensity at a later time $I(t+\tau)$:

$$G_2(\tau) = \langle I(t)I(t+\tau)\rangle, \tag{1.16}$$

where τ is the lag time between points. For a beam with intensity $I(t)$, Eq. 1.16 is normalized to

$$g_2(\tau) = \frac{\langle I(t)I(t+\tau)\rangle}{\langle I(t)\rangle^2}. \tag{1.17}$$

The normalized intensity autocorrelation function measures the similarity between the scattered intensity at time t and at a later time $t+\tau$. For particles undergoing Brownian motion, this similarity decreases with increasing lag time because the speckle pattern changes as the particles diffuse.

The first-order correlation function $G_1(\tau)$ correlates the scattered electric fields rather than the intensities. For scattered electric fields $E(t)$ and $E(t+\tau)$,

$$G_1(\tau) = \langle E^*(t)E(t+\tau)\rangle. \tag{1.18}$$

Following a similar procedure, the normalized function $g_1(\tau)$ is written as

$$g_1(\tau) = \frac{\langle E^*(t)E(t+\tau)\rangle}{\langle |E(t)|^2\rangle}. \tag{1.19}$$

Equations 1.17 and 1.19 are related through the Siegert relation [6], which applies when the detected field is a Gaussian random process arising from many independent scatterers:

$$g_2(\tau) = 1 + \beta|g_1(\tau)|^2, \tag{1.20}$$

where β is the coherence factor determined by optical alignment, detector size, and collection geometry.

When a sample contains particles of a single size, it is defined to be monodisperse. For a monodisperse suspension undergoing Brownian motion, the field correlation decays as a single exponential,

$$g_1(\tau) = e^{-\tau\Gamma}, \tag{1.21}$$

where Γ is defined by Eq. 1.1. Combining Eqs. 1.20 and 1.21 gives

$$g_2(\tau) = 1 + \beta e^{-2\tau\Gamma}. \tag{1.22}$$

In a polydisperse system a single exponential is only an effective description; a more complete model integrates over a distribution of decay rates. Substituting Eq. 1.1 into Eq. 1.22 yields

$$g_2(\tau) = 1 + \beta e^{-2\tau D q^2}. \tag{1.23}$$

The Stokes–Einstein diffusion coefficient is

$$D = \frac{k_B T}{6\pi\eta R_H}, \tag{1.24}$$

where k_B is Boltzmann's constant, T is the absolute temperature, η is the solvent viscosity, and R_H is the hydrodynamic radius. The hydrodynamic radius accounts for the size of the particle and the shape it takes when moving through a liquid. For the latex standards, this includes a pocket of air surrounding the spheres.

Using the autocorrelation function to analyze scattered light intensity timeseries should resolve the diameter of the particles emitting the scattered light. This type of analysis is commonly done within dynamic light scattering experiments, but can be applied to other techniques. The following section introduces dynamic light scattering, its limitations, and some of the industrial applications.

1.5 Dynamic light scattering

The particles used for DLS experiments are typically on or near the nanometer scale. Particles with diameters less than about 100 nm are not visible with conventional

optical microscopes and require other methods such as electron microscopy or DLS. Particles at this scale have a high surface-area-to-volume ratio, which is important in biology, chemistry, and related applications [7].

DLS is a non-invasive technique that probes samples in their native state. In biology, the study of macromolecules by DLS can reveal the homogeneity of proteins and related complexes [8, 9]. Understanding these macromolecules is important for describing how proteins bind, transport molecules, and catalyze chemical reactions. The technique has also been applied to more complex systems, including emulsions and food products, where particle-size measurements can provide insight into processing and stability. These applications demonstrate the broad utility of DLS for characterizing particle systems, while also highlighting the need to understand the limitations of the technique when analyzing complex or polydisperse samples [10, 11].

Although DLS can provide a low-cost, fast, and non-destructive estimate of the hydrodynamic diameter of a nanoparticle [8], it has limitations. DLS assumes that the particles under study are spherical. A rod or irregularly shaped particle may therefore be reported as an equivalent sphere whose diameter does not fully represent the sample. The technique also has limited ability to resolve particles that do not differ greatly in size; if two populations do not differ by roughly a factor of three, they may not be distinguished. DLS is highly sensitive to sample concentration: a dilute sample produces a weak signal, while an overly concentrated sample can lead to multiple scattering. The sample refractive index and temperature must also be known accurately to calculate particle size. Despite these limitations, DLS remains a valuable technique for characterizing particles or macromolecules under native conditions.

1.6 Motor control and diffractometer

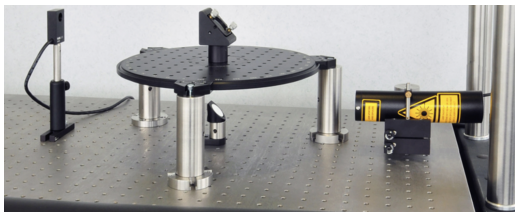
The experimental procedure for conducting multi-angle DLS was refined in order to allow for controlled angular changes of the detector while the sample remained undisturbed. This section describes the building of the experimental apparatus that allowed for quantitative multi-angle DLS experiments.

A motorized diffractometer was incorporated into the experimental apparatus to provide precise and repeatable positioning of the detector during multi-angle DLS measurements. The rotating breadboard, shown in Fig. 1.6, was used as the diffractometer platform, allowing the detector to move along a circular arc while the sample remains at the center of rotation. This geometry permits measurements at different scattering angles while observing the same scattering volume, reducing alignment errors between successive measurements.

The diffractometer was driven by a belt-coupled stepper motor using a custom 3D-printed GT2 gear mounted directly to the rotating breadboard. As shown in Fig. 1.7, spacers were used to provide sufficient clearance for the drive system while maintaining a rigid mechanical connection. The belt was tensioned using guide bearings to minimize backlash and improve the repeatability of angular positioning.



(a) A Thorlabs large-area rotating breadboard configured as a diffractometer.



(b) Example mounting arrangement showing a laser beam passing through the central aperture of the rotating breadboard.

Figure 1.6: Commercial large-area rotating breadboard used as the basis for the motorized diffractometer (Fig. 1.6a; mounting example in Fig. 1.6b). Images courtesy of Thorlabs.

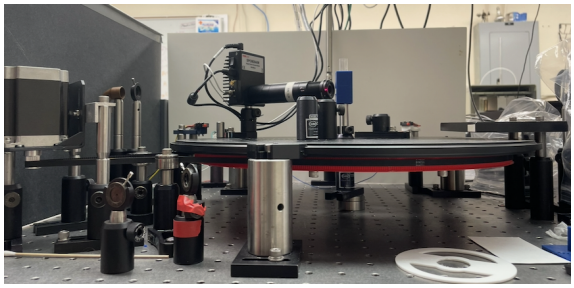


Figure 1.7: Motorized diffractometer used in the experimental apparatus. Aluminum spacers provide clearance between the rotating breadboard and the optical table for the drive gear. The GT2 gear is mounted beneath the breadboard and is driven by the stepper motor shown on the left.

Detector motion was controlled through software commands issued by the correlation system and interpreted by a Raspberry Pi-based controller. Once calibrated, the diffractometer provided reproducible angular positioning with a minimum step size of approximately 0.25° and an estimated angular uncertainty of ±0.08°. Measurements were typically acquired using incremental angular changes of no more than 20° to preserve alignment and minimize the accumulation of positioning errors. Limit switches established a reproducible reference position before each measurement sequence, ensuring that detector angles remained consistent throughout the experiment.

Figure 1.8 shows the completed diffractometer with the sample located at the center of rotation. The incident laser beam enters the cuvette from below, while the detector is translated along the circular path to collect scattered light at different angles. Because the sample remains at the center of rotation, changing the detector angle does not alter the scattering volume being observed. This geometry improves the repeatability of multi-angle measurements by maintaining a consistent optical alignment throughout the experiment. The definition of θ in this apparatus is the same as in Fig. 1.2.

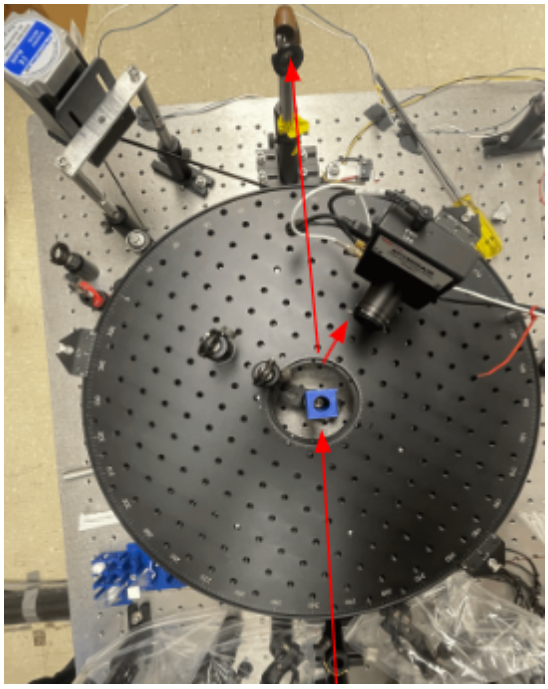


Figure 1.8: Motorized diffractometer used for multi-angle measurements. The stepper motor drives the rotating breadboard through a belt and gear system. The sample is positioned at the center of rotation, allowing the detector to be moved to different scattering angles while maintaining observation of the same scattering volume. The short red arrow illustrates one possible scattered-light path collected by the detector, while the longer arrow indicates the transmitted beam.

1.7 Custom 3D-printed optomechanical parts

Building a multi-angle DLS bench from commercial optical hardware still leaves interfaces that stock mounts do not provide: seating a cylindrical cuvette on the center of rotation, clearing laser and detector lines of sight at many angles, homing a belt-driven stage, and adapting a nonstandard diode laser to one-inch optic hardware. Those interfaces were filled with custom fused-deposition-modeled (FDM) parts designed for this apparatus, which make the geometry of Section 1.6 practical on the table.

Seven part families were designed and iterated during construction (Fig. 1.9): a

stiffened limit-switch mount for homing the diffractometer (Fig. A.1); a centering cap that locates a sample post in the Thorlabs stage aperture (Fig. A.2); a cuvette stage with beam and scattering windows (Fig. A.3); an alignment pin that occupies the cuvette seat and scatters the laser for beam centering (Fig. A.4); a numbered cuvette organizer for sample preparation (Fig. A.5); a two-piece laser adaptor for one-inch optic mounts (Fig. A.6); and a large GT2 ring gear that bolts under the rotating breadboard and engages a timing belt from the stepper motor (Fig. A.7). Several designs went through multiple revisions. Appendix A catalogs each part with photographs and STL renderings where available.

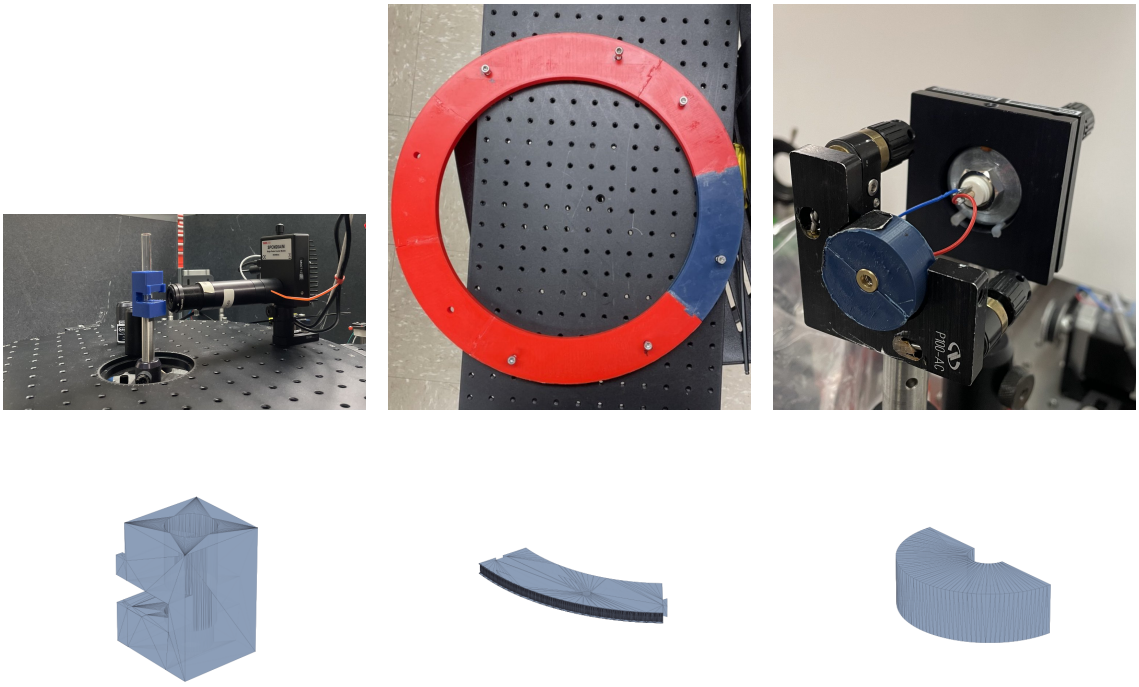


Figure 1.9: Representative custom prints central to the multi-angle geometry. Left: cylindrical cuvette stage (photo) and STL rendering. Center: assembled GT2 diffractometer gear (photo) and one printable gear segment (STL). Right: diode-laser adaptor in a one-inch optic mount (photo) and STL half. Additional parts and detail appear in Appendix A.

Together they center the scattering volume, set a reproducible angular origin, trans-

mit stepper motion to the rotating stage, and hold the diode laser in a standard optic mount.

CHAPTER 2

Experiment

This chapter describes the experimental procedures used for the multi-angle DLS measurements: choosing an operating concentration for each latex standard, following a standard sample-preparation protocol, and characterizing the laser wavelength used to compute q. The calibrated multi-angle results themselves appear in Chapter 3.

2.1 Experimental determination of optimal concentration

An important parameter in dynamic light scattering (DLS) experiments is the sample concentration. At high concentrations, photons may undergo multiple scattering, in which an incident photon is scattered by one particle and then re-scattered by another. This effect distorts the measured autocorrelation function and must be minimized (Fig. 2.1).

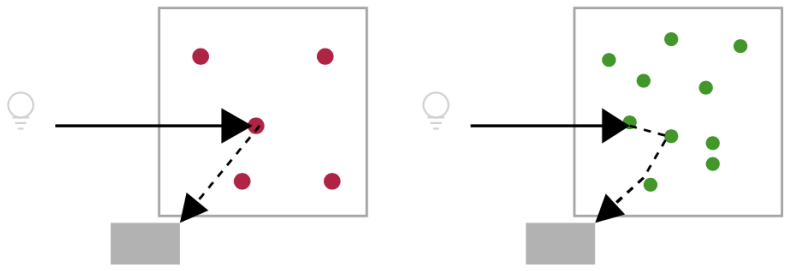


Figure 2.1: An illustration of multiple scattering. A light bulb emits light. The emitted light is scattered by particles before being collected by the detector (gray box). On the right, over-concentration leads to multiple scattering and incorrect estimates of particle size.

Three latex sphere sizes were purchased from Ted Pella as a calibration set, which

included 0.09 μm, 0.30 μm, and 0.49 μm latex spheres. Before the multi-angle calibration measurements in Chapter 3, a qualitative dilution screen was used to choose an operating concentration for each size. Samples were prepared by diluting manufacturer stock into distilled water and judged by count rate and curve shape (noise versus multiple-scattering distortion), not by extracting sizes as a function of concentration.

Based on previous measurements, the useful avalanche photodiode detector (APD) count-rate window was taken to lie between 4.0×10^5 and 6.0×10^5 counts per second, particularly at the largest scattering angle measured. An initial sample of known concentration was therefore used to establish the appropriate range for the dilution screen.

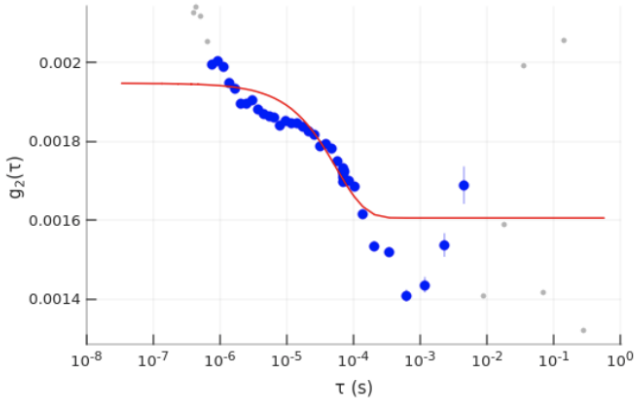
Three samples were prepared using distilled water and latex spheres of varying diameters (0.09 μm, 0.30 μm, and 0.49 μm), each at a concentration of 0.01%. This concentration was achieved by combining 10 μL of latex stock with 990 μL of distilled water using a micropipette.

2.2 Dilution screen for operating concentration

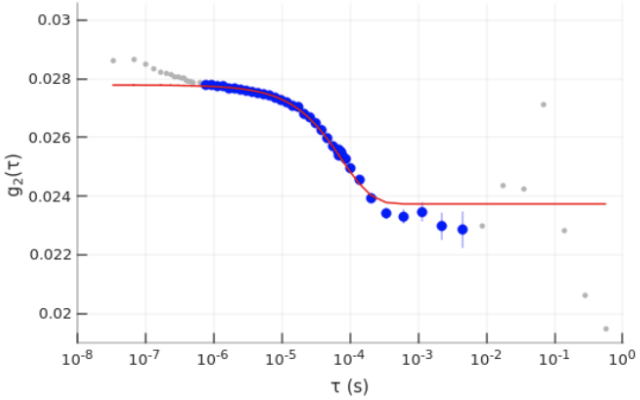
Figure 2.2 shows representative autocorrelation functions for the 0.09 μm latex only; analogous screens for 0.30 μm and 0.49 μm are summarized in the text and in Table 2.1. The graphs in Fig. 2.2 (a)-(c) demonstrate that low concentrations are dominated by noise, while higher concentrations show multiple-scattering distortion.

The 0.09 μm sample at 0.01% solids produced a count rate below 5.0×10^3 counts per second, which was insufficient for reliable measurements (Fig. 2.2a). Raising the solids concentration into the 0.10–0.20% range improved the count rate. Concentrations near 0.20% produced multiple-scattering effects (Fig. 2.2c), while concentrations near 0.10–0.15% produced sufficient counts and the expected single-scattering

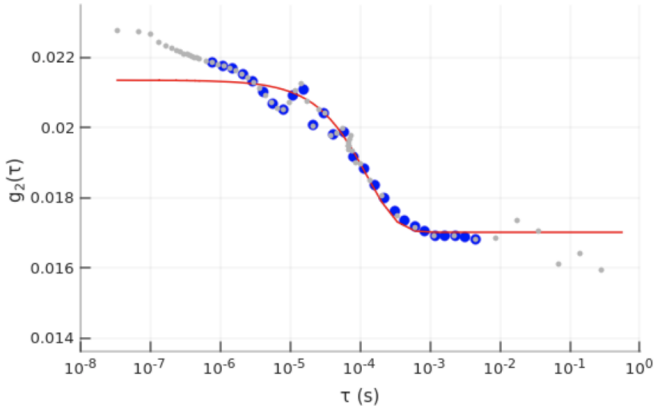
sigmoid shape (Fig. 2.2b). Concentrations quoted here are final solids percentages in the prepared cuvette sample, not stock-solution volume fractions.



(a) A 0.09 μm latex sample at 0.01% solids, dominated by noise.



(b) A 0.09 μm latex sample at 0.10% solids.



(c) A 0.09 μm latex sample at 0.20% solids, showing multiple-scattering distortion.

Figure 2.2: Representative intensity autocorrelation functions $g_2(\tau)$ at three solids concentrations for 0.09 μm latex. Only this size is shown graphically; the same qualitative criteria were used to select operating concentrations for 0.30 μm and 0.49 μm latex (Table 2.1).

For the 0.30 μm sample, 0.01% solids produced an autocorrelation shape consistent with multiple scattering. The usable range was approximately 0.001–0.0025%, with 0.002% giving the most stable single-scattering appearance.

For the 0.49 μm sample, concentrations near 0.0005% provided the most stable single-scattering appearance; higher concentrations showed multiple-scattering signatures.

Particle Size (μm)	Usable Range (%)	Selected (%)
0.09	0.10 – 0.15	0.15
0.30	0.001 – 0.0025	0.002
0.49	$\sim$0.0005	0.0005

Table 2.1: Operating concentrations selected from the dilution screen and used for the multi-angle calibration measurements in Chapter 3.

The multi-angle latex results in Chapter 3 were acquired at these selected concentrations.

2.3 Standard operating procedure

A low-volume Type 76 light-scattering fluorescence cuvette from FireflySci was used, with an outer diameter of 12 mm and a light path of 10 mm. The same procedure can be adapted to other DLS test benches.

Cuvette cleaning

- Soak the cuvette in a 1:1 mixture of ethanol and distilled water for 2 minutes to remove deposits.
- Thoroughly rinse the cuvette with distilled water.
- Dry the cuvette.
- Visually inspect the cuvette for dust or remaining deposits.

The samples were prepared from stock solutions of polystyrene latex spheres. These stock solutions are monodisperse and require care to avoid contamination. The manufacturer-specified size uniformity, reported as a coefficient of variation (C.V.), is listed in Table 2.2.

Particle Size (μm)	Size uniformity (C.V.)
0.09	$\leq 15\%$
0.30	$\leq 3\%$
0.49	$\leq 3\%$

Table 2.2: Manufacturer-specified size uniformity (coefficient of variation of the particle-size distribution), not an uncertainty on the mean diameter.

Proper sample preparation and handling are critical for obtaining reliable DLS measurements. Latex nanosphere suspensions were prepared using distilled water, with careful attention given to minimizing contamination, bubbles, and particle aggregation. Following the manufacturer's recommendation, the stock solution was stored under temperature-controlled conditions to maintain stability. The following standard operating procedure was used to prepare samples for testing and validating quantitative measurements with the FPGA correlator.

- Clean the cuvette as described under "Cuvette cleaning" above.
- Fill a 2 mL capsule with distilled water. Remove the stock solution from the refrigerator.
- Identify the final solids concentration being prepared (e.g. 0.15% solids, obtained by diluting the manufacturer stock into distilled water to a total volume of 1 mL).
- Rotate the stock solution slowly to mix without aggregating the particles.
- Dispense the appropriate volumes of stock solution and water into a labeled 2 mL capsule.
- Close the capsule lid and rotate it to mix without introducing air bubbles.

- Power on the laser, avalanche photodiode, motor drive, and FPGA autocorrelation device.
- Dispense the sample into the dry cuvette and place it into the cuvette mount at the center of the test bench.
- Enter the desired angles of examination and a detailed file name before starting each experiment.

Additional precautions, including the use of gloves and lint-free materials, were implemented to minimize contamination from dust and oils. These procedures helped ensure that measured scattering signals were representative of the sample rather than experimental artifacts. Following the manufacturer's suggestion, a 1:1 mixture of ethanol and distilled water was used to remove deposits. Ethanol was found to be important for maintaining a clear quartz cuvette.

Cleaning procedures were refined over the course of the experiments as measurements at different scattering angles revealed features in the autocorrelation function that suggested surface deposits on the cuvette (Fig. 2.3).

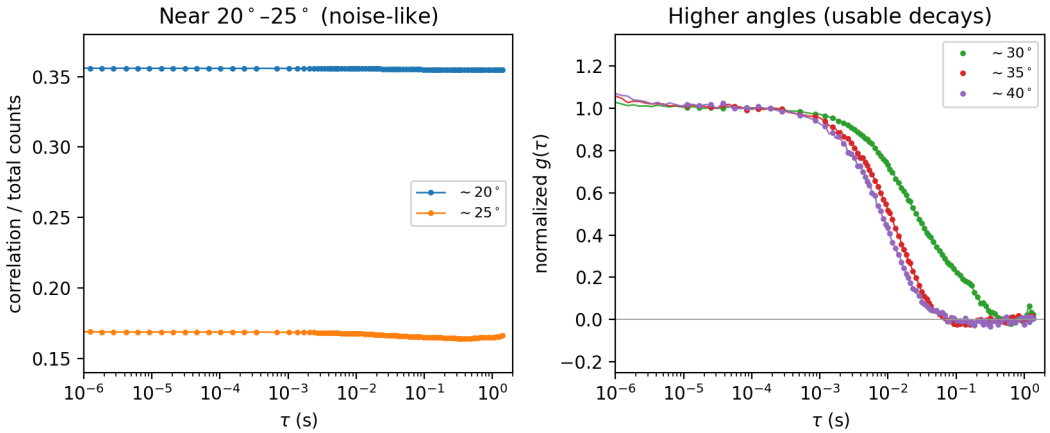


Figure 2.3: Left: angles near 20° and 25° give nearly flat autocorrelation functions (noise-like), attributed to surface deposits on the cuvette. Right: higher angles from the same run show usable decays after display normalization.

To evaluate whether these features originated from the sample, the cuvette, or contamination, the cuvette was rotated without disturbing the contents. The measurement was then repeated and compared with Fig. 2.3 to determine whether the locations of these features had changed. When a brief rotation changed the autocorrelation function, a brief soak removed residual deposits before subsequent measurements were performed.

2.4 Laser characteristics

A red diode laser was used as the light source. The laser was repurposed from a commercial device because it could be controlled electronically, enabling future intensity modulation.

The peak emission wavelength was measured with an Ocean Optics USB 2000 grating spectrometer as a function of drive voltage (Fig. 2.4). At the operating voltage used for scattering ($\approx$ 2.7–3.1 V) the peak is 653 nm; above about 4 V the laser shifts to longer wavelengths. In the multi-angle analysis a value of 653 nm was used for the scattering vector; the $\pm$2 nm wavelength prior in Section 4.3 covers the variation from the 653 nm operating peak. This spectrum was also used to select an optical bandpass filter in front of the detector to reject background light.

With the concentration screen, sample protocol, and laser wavelength established, the next chapter presents the correlator timing tests and the multi-angle scattering measurements.

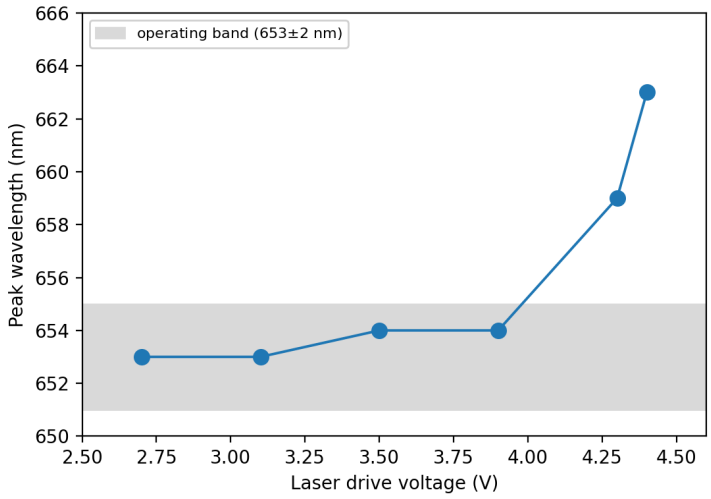


Figure 2.4: Peak emission wavelength versus laser drive voltage. The shaded band marks the 653 ± 2 nm operating region used for the DLS measurements.

CHAPTER 3

Results

This chapter reports the measurements in the order they support the validation argument: electronic square-wave timing tests of the FPGA correlator; multi-angle latex-sphere autocorrelations and Γ versus q^2 fits; isolation of motor-induced noise; and a qualitative coffee-creamer test on a nonideal sample. Interpretation of the recovered diameters and the uncertainty budget is deferred to Chapter 4.

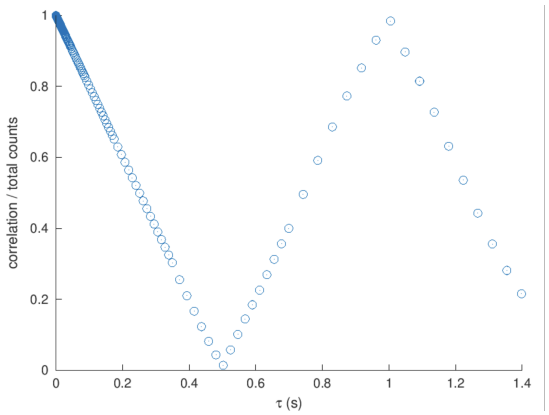
3.1 Correlation of a test waveform using square waves

The FPGA correlator was first tested using periodic electronic square-wave signals with known frequencies. These measurements provided a controlled test of the correlator before applying it to photon-counting DLS data. A square wave has a periodic autocorrelation function whose peaks are separated by the period of the input waveform. Therefore, a 1 Hz signal should produce correlation peaks separated by 1 s, a 100 Hz signal by 10^{-2} s, and a 10 kHz signal by 10^{-4} s. Agreement between the measured peak spacing and the known input period verifies the timing response of the correlator.

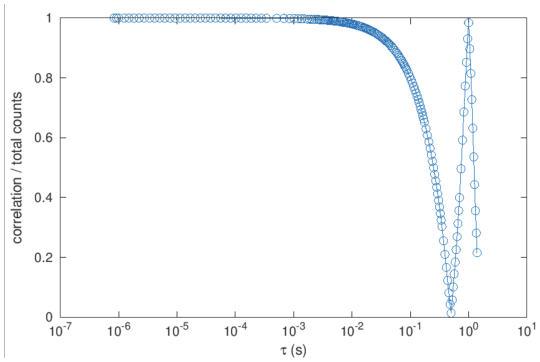
Square waves do not reproduce the stochastic statistics of photon arrivals in a DLS experiment, but they provide a reproducible timing standard for verifying the response of the correlator. The autocorrelation function of a square wave has the shape of a triangle or ramp function. Although these autocorrelation functions are commonly viewed on a linear scale, the same data are also shown on a logarithmic scale to highlight the range of lag times that the FPGA-based correlator can process.

To evaluate the performance limits of the FPGA correlator and its ability to resolve rapidly varying signals, square-wave inputs with frequencies of 1 Hz (Fig. 3.1), 100 Hz (Fig. 3.2), and 10 kHz (Fig. 3.3) were analyzed. Data points represent the calculated autocorrelation values at discrete lag times τ. The lag-time sampling is nonuniform, providing finer resolution at short lag times where the autocorrelation changes rapidly and progressively larger intervals at longer lag times. For small values of τ, the autocorrelation function remains close to 1 because the signal is highly correlated with itself. As the lag time increases, the autocorrelation decreases until the shifted waveform again overlaps with the original square wave, producing the periodic structure observed in the autocorrelation function.

For a 1 Hz waveform, the spacing between rising edges is 1 s. This is evident on a linear scale (Fig. 3.1a) and on a logarithmic scale (Fig. 3.1b), with a peak in the autocorrelation function at 1 s.



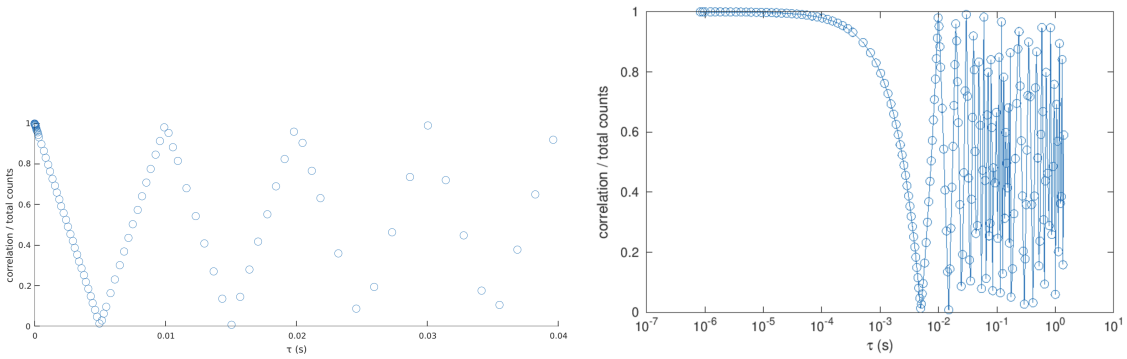
(a) An autocorrelation plot of a 1 Hz square wave on a linear scale.



(b) An autocorrelation plot of a 1 Hz square wave on a logarithmic scale.

Figure 3.1: Autocorrelation of a 1 Hz square-wave input shown on linear and logarithmic lag-time axes. The measured correlation peak occurs at approximately 1 s, matching the period of the input waveform.

For a 100 Hz waveform, the spacing between rising edges decreases. On a linear scale (Fig. 3.2a) and on a logarithmic scale (Fig. 3.2b) the peak in the autocorrelation function occurs at 10^{-2} s.



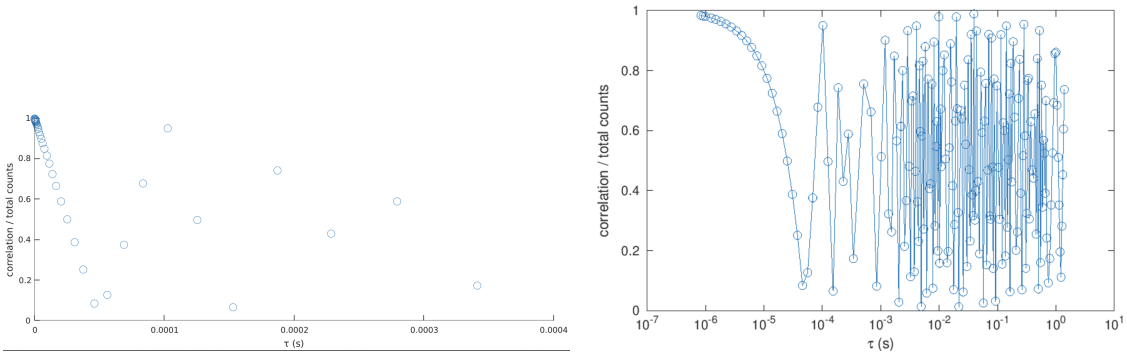
(a) An autocorrelation plot of a 100 Hz square wave on a linear scale.

(b) An autocorrelation plot of a 100 Hz square wave on a logarithmic scale.

Figure 3.2: Autocorrelation of a 100 Hz square-wave input. The first repeated correlation peak occurs near 10^{-2} s, consistent with the 0.01 s input period.

To investigate the upper frequency limit at which the FPGA correlator can recover a known *periodic* electronic signal, a 10 kHz square wave was used as input. The characteristic triangular autocorrelation function remains clearly visible in Fig. 3.3a. Figure 3.3b shows the same data on a logarithmic time scale. The first correlation peak occurs at approximately 10^{-4} s, corresponding to the 100 μs period of the 10 kHz square wave. On the log plot, the dense oscillations beyond about 10^{-3} s are the same 100 μs periodicity undersampled by the coarser multi-tau lag spacing at long delays, not a correlator failure.

Exploratory square-wave tests near 100 kHz did not recover a clean ladder of peaks at the expected 10 μs spacing. That result bounds the present periodic timing check: it does not mean the correlator only operates up to 10 kHz. Photon correlation still registers short coincidence windows (tens of nanoseconds in the multi-tau lag bank) and long-lived correlations over the acquisition. The DLS decays reported later sit on millisecond timescales, well away from that 100 kHz periodic test. The 10 kHz square-wave data are therefore treated as a successful periodic verification, and the 100 kHz attempt only as a limit of that particular electronic check.



(a) An autocorrelation plot of a 10 kHz square wave on a linear scale.

(b) An autocorrelation plot of a 10 kHz square wave on a logarithmic scale.

Figure 3.3: Autocorrelation of a 10 kHz square-wave input. The correlation peak near 10^{-4} s demonstrates that the FPGA correlator resolves periodic signals on sub-millisecond timescales.

These square-wave results confirm the correlator's timing response before turning to photon-counting DLS data.

3.2 Correlation and hydrodynamic diameter of latex spheres

After the electronic timing tests, the FPGA correlator was evaluated using DLS measurements from calibrated latex nanosphere suspensions. These measurements provide the primary validation of the apparatus: for particles undergoing Brownian diffusion, the field decay rate Γ must increase linearly with q^2. For each particle size, intensity autocorrelation functions were recorded at several detector angles. The FPGA accumulates coincidence counts $C(\tau)$ in each lag channel—the number of times a photon arrival is accompanied by another arrival delayed by τ. Over the same run it also records N, the total number of photons detected during the acquisition. The interface computer then normalizes the lag channels by that acquisition total,

$$g(\tau) = \frac{C(\tau)}{N}. \tag{3.1}$$

This $g(\tau)$ is proportional to the normalized intensity autocorrelation $g_2(\tau)$ of Chapter 1, but its absolute vertical scale is set by the optical collection geometry and by residual uncorrelated light. At long lag times the measured curve therefore levels off at some baseline value rather than at a textbook 0 or 1. In the fit that baseline is treated as a free parameter B, so that Γ is determined by how fast the excess above the baseline decays, not by the absolute height of the curve. The multi-angle ACF figures plot a display-normalized form $(g - B)/\beta$ that runs from near 1 to near 0, only for visual comparison across angles.

The per-lag uncertainty used for weighting follows from treating the coincidence tally C as a Poisson count. At fixed rate, a count of C coincidences has variance C, so $\sigma_C = \sqrt{C}$. The quantity that enters the fit is the normalized ratio $g = C/N$. Here N is the acquisition total, a single large number measured once for the run; relative to any one lag channel it is essentially fixed, so dividing by N scales both the value and its uncertainty:

$$\sigma_g(\tau) = \frac{\sigma_C}{N} = \frac{\sqrt{C(\tau)}}{N} = \sqrt{\frac{g(\tau)}{N}}. \tag{3.2}$$

That is why the bars are neither $\sqrt{N}$ nor $\sqrt{C}$ alone: $\sqrt{N}$ would describe fluctuations of the acquisition total itself, and $\sqrt{C}$ would be the uncertainty on the raw coincidence count before normalization. Because the fit uses $g = C/N$, the coincidence error must be divided by N, which is the same as $\sqrt{g/N}$. Although N is the same at every lag, C (and therefore σ_g) still varies with τ. Each curve was then fit by weighted least squares with weights $1/\sigma_g^2$ to

$$g(\tau) = B + \beta\, e^{-2\Gamma\tau}, \tag{3.3}$$

where B is the fitted baseline, β is the optical coherence (contrast) factor, and Γ is the field-correlation decay rate. Points with larger σ_g therefore contribute less to the determination of Γ than points with smaller σ_g. Short lag times below 1 μs were excluded from the fits (e.g. Fig. 3.4). At those delays the measured $g(\tau)$ often rises above the optical contrast plateau in a manner characteristic of detector

afterpulsing rather than Brownian decay: a spurious secondary pulse follows a real photon detection and inflates the short-lag coincidence rate. Because the particle correlation times of interest here are much longer—of order milliseconds for the latex standards under the measured angles—the afterpulsing feature lies well outside the physically informative part of the curve and omitting it does not change the extracted Γ at the precision of these fits. Where the residuals about the fit exceed the Poisson σ_g, the reported uncertainties on Γ are enlarged using the reduced χ^2 of the fit so that they reflect the observed scatter. The resulting Γ values were plotted against q^2 and fit to $\Gamma = Dq^2 + b$ with weights $1/\sigma_\Gamma^2$. A free intercept b is allowed so that a small constant offset from residual baseline or angle-calibration error does not force bias into the slope D. A linear relationship with positive slope indicates that the measured fluctuations arise from diffusive particle motion rather than only instrumental artifacts.

Because the diffusion coefficient decreases with increasing particle radius, smaller latex spheres are expected to decorrelate faster than larger spheres at the same scattering angle. The measured autocorrelation curves should therefore shift toward longer decay times as the particle diameter increases. The coherence factor β is set by optical alignment, detector size, and collection geometry.

Three suspensions of latex spheres with different particle diameters were measured at the operating concentrations selected in Table 2.1, and the recovered hydrodynamic diameters were compared with the manufacturer's specified diameters (Table 3.1). All measurements were performed over an angular range of 32.5°–44.5° with an angular step size of 2°. At the largest angle the recorded APD count rates fell in or near the 4–6 $\times$ 10^5 s^{-1} window targeted in Chapter 2.

Figure 3.4 shows the normalized autocorrelation functions for all angles for 90 nm latex spheres at 0.15% solids. As photon arrival events are detected, the FPGA updates the autocorrelation function in real time. Larger scattering angles decay more rapidly, as expected from the q^2 dependence of Γ.

Table 3.1: Comparison of nominal and recovered hydrodynamic diameters from single-exponential multi-angle fits at the concentrations of Table 2.1. Uncertainties are the combined fit-and-instrumental budget of Section 4.3. All measurements used the angular range 32.5°–44.5° with a 2° step.

Particle size (μm)	Conc. (%)	Extracted diameter (μm)	Difference (%)
0.09	0.15	0.094 ± 0.008	4
0.30	0.002	0.397 ± 0.028	32
0.49	0.0005	0.556 ± 0.064	14

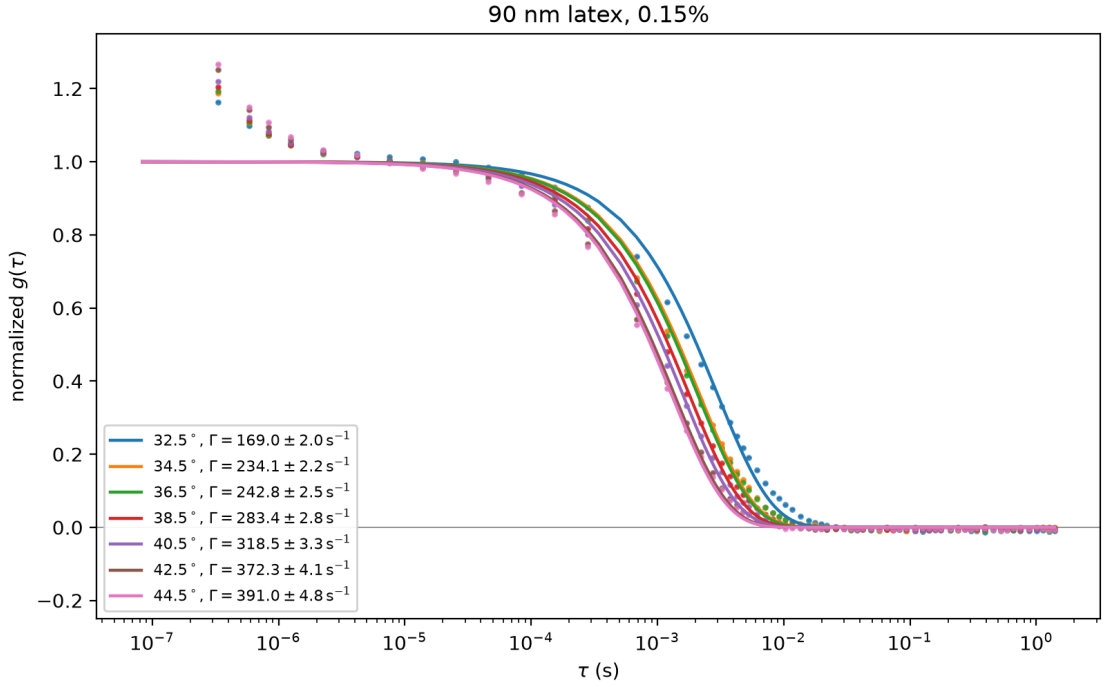


Figure 3.4: Multi-angle intensity autocorrelation functions for 90 nm latex spheres in distilled water at 0.15% solids. Solid curves are single-exponential fits weighted by analytic Poisson uncertainties $\sigma_g = \sqrt{g/N}$. The upturn at $\tau \lesssim 1$ μs is attributed to detector afterpulsing and is excluded from the fits; the Brownian decays themselves occur on much longer (millisecond) timescales. Larger scattering angles decay more rapidly, consistent with $\Gamma \propto q^2$.

Plotting the fitted Γ against q^2 (Eq. 1.1) should be linear with slope equal to the diffusion coefficient. For the 90 nm sample, Figure 3.5 gives $D = 5.24 \times 10^{-12}$ m^2/s ($R^2 = 0.977$) and a hydrodynamic diameter of 94 ± 8 nm.

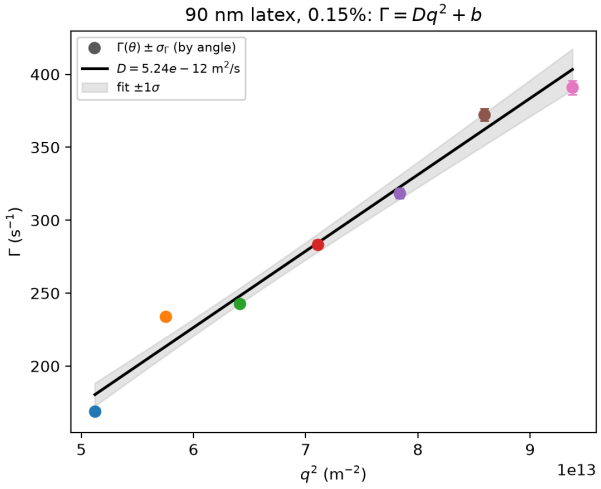


Figure 3.5: Decay rate Γ from weighted single-exponential fits as a function of q^2 for 90 nm latex spheres. Vertical bars are $\pm 1\sigma$ uncertainties from the weighted curve fits; the shaded band is the $\pm 1\sigma$ uncertainty of the weighted linear fit. The slope is the diffusion coefficient used to obtain the hydrodynamic diameter.

For the 300 nm latex spheres, a concentration of 0.002% provided adequate scattered photons at all measured angles (Fig. 3.6). Each angle was fit independently for Γ.

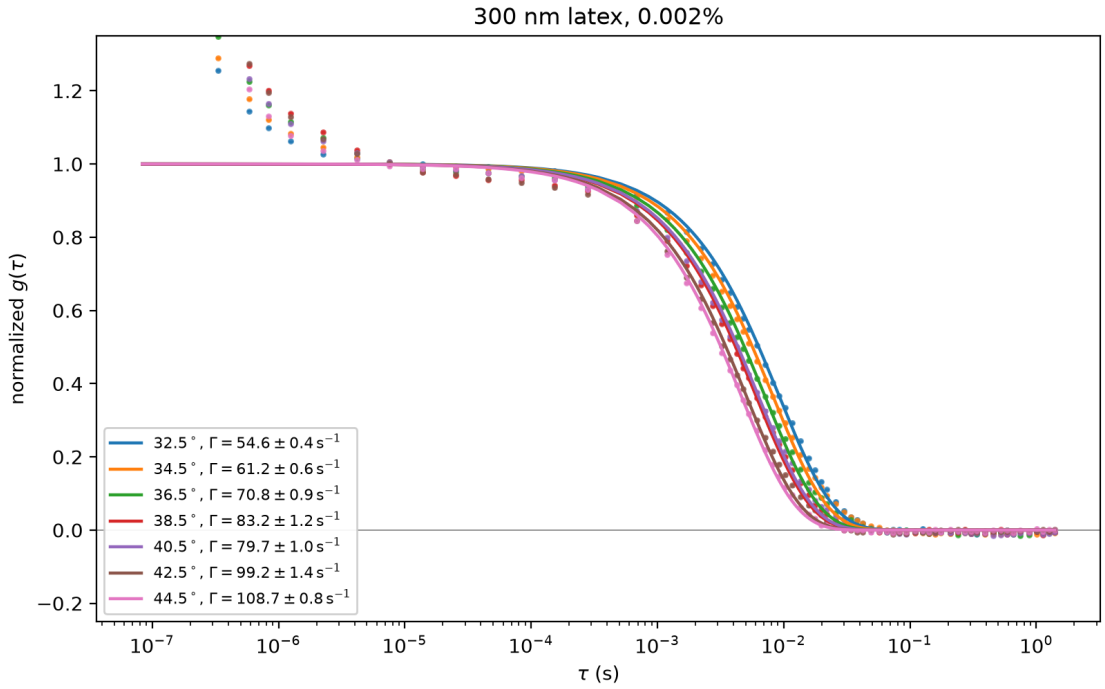


Figure 3.6: Multi-angle intensity autocorrelation functions for 300 nm latex spheres in distilled water at 0.002% solids, with single-exponential fits weighted by analytic Poisson uncertainties $\sigma_g = \sqrt{g/N}$. As in Fig. 3.4, the short-lag upturn ($\tau \lesssim 1$ μs) is attributed to detector afterpulsing and is excluded from the fits.

Figure 3.7 shows Γ versus q^2 for the 300 nm sample. The weighted linear fit gives $D = 1.23 \times 10^{-12}$ m^2/s ($R^2 = 0.959$) and a hydrodynamic diameter of 397 ± 28 nm.

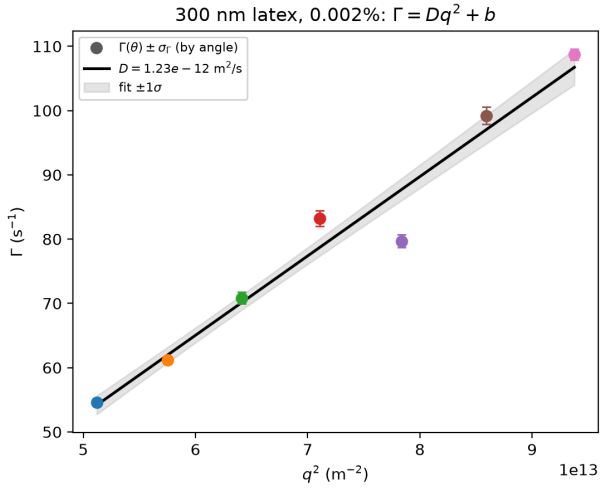


Figure 3.7: Decay rate Γ versus q^2 for 300 nm latex spheres, with $\pm 1\sigma$ weighted-fit point uncertainties and a $\pm 1\sigma$ linear-fit band. The fitted slope is the diffusion coefficient.

A 0.0005% sample of 490 nm latex spheres was measured in the same way (Fig. 3.8). Larger particles decorrelate more slowly at a given angle, as expected from the smaller diffusion coefficient.

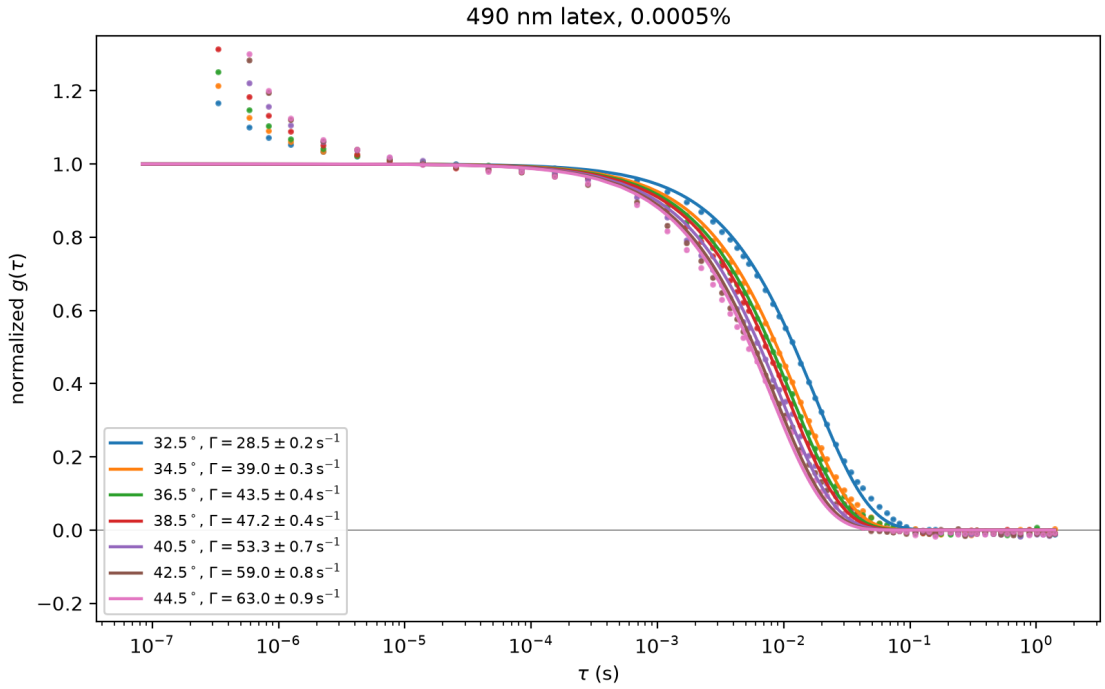


Figure 3.8: Multi-angle intensity autocorrelation functions for 490 nm latex spheres in distilled water at 0.0005% solids, with single-exponential fits weighted by analytic Poisson uncertainties $\sigma_g = \sqrt{g/N}$. Short-lag points below 1 μs are excluded from the fits as detector afterpulsing (see Fig. 3.4).

Figure 3.9 yields $D = 8.82 \times 10^{-13}$ m^2/s ($R^2 = 0.949$) and a hydrodynamic diameter of 556 ± 64 nm.

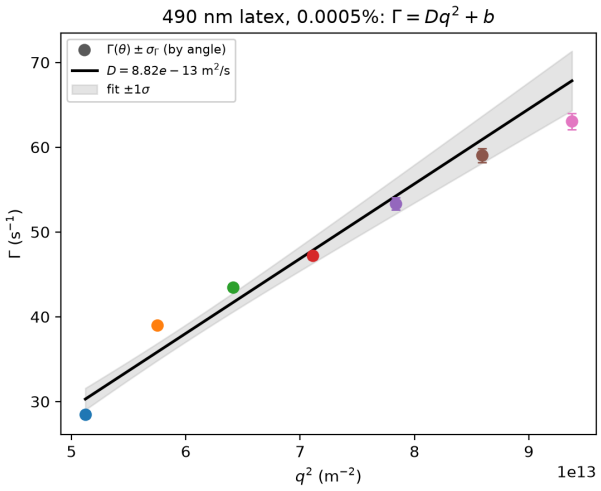


Figure 3.9: Decay rate Γ versus q^2 for 490 nm latex spheres, with $\pm 1\sigma$ weighted-fit point uncertainties and a $\pm 1\sigma$ linear-fit band. The fitted slope gives the diffusion coefficient and corresponding hydrodynamic diameter.

3.3 Isolating and eliminating motor-induced noise

During multi-angle measurements, the motorized diffractometer introduced an additional instrumental artifact. When the stepper motor remained energized during data collection, the autocorrelation curves showed visible jitter, particularly near the plateau and baseline regions (Fig. 3.10). Disabling the motor after each angular move substantially reduced this structure (Fig. 3.11). This observation suggests that the motor system introduced a measurable artifact, likely through electrical pickup, mechanical vibration, or both, and that the FPGA correlator is sensitive enough to reveal it.

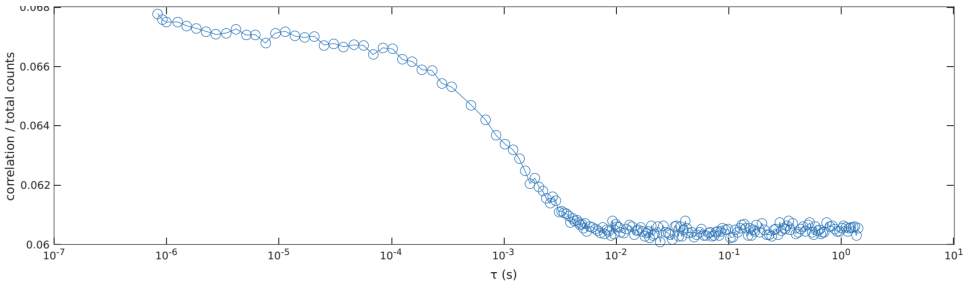


Figure 3.10: Autocorrelation function measured from a 90 nm latex-sphere sample with the stepper motor energized during data collection. The curve exhibits visible jitter, indicating a motor-induced artifact.

In Figure 3.11, a single autocorrelation function was recorded for a 90 nm sample at 0.15% solids with the motor disabled after an angular change. No continuous drive signals were then sent to the stepper motor. The jitter present when the motor is energized is less apparent when the motor is not energized after angular changes.

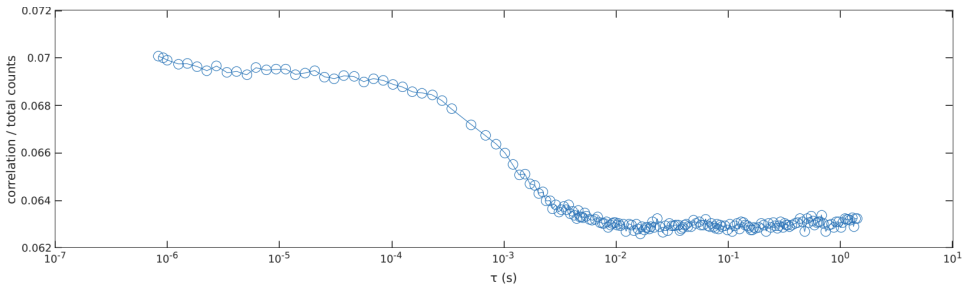


Figure 3.11: Autocorrelation function measured from the same 90 nm latex-sphere sample after disabling the stepper motor during data collection. The reduced jitter indicates improved measurement stability.

Although this noise was introduced during installation of the test bench, it was isolated and then removed from the acquisition procedure to improve the signal-to-noise ratio.

3.4 Correlation of coffee creamer

Following validation with calibrated latex standards, a diluted coffee-creamer sample was measured as a qualitative test on a nonideal scatterer. Unlike the latex standards, coffee creamer is polydisperse and does not have a single well-defined particle diameter. Four angles from 25.5° to 40.5° were used rather than the seven-angle latex protocol. Each angle was fit to the same weighted single-exponential model used for the latex standards. Figure 3.12 shows the normalized autocorrelation functions, and Figure 3.13 shows Γ versus q^2. The slope corresponds to an effective hydrodynamic diameter of approximately 560 nm under a single-exponential model; it is not a unique particle size for the suspension.

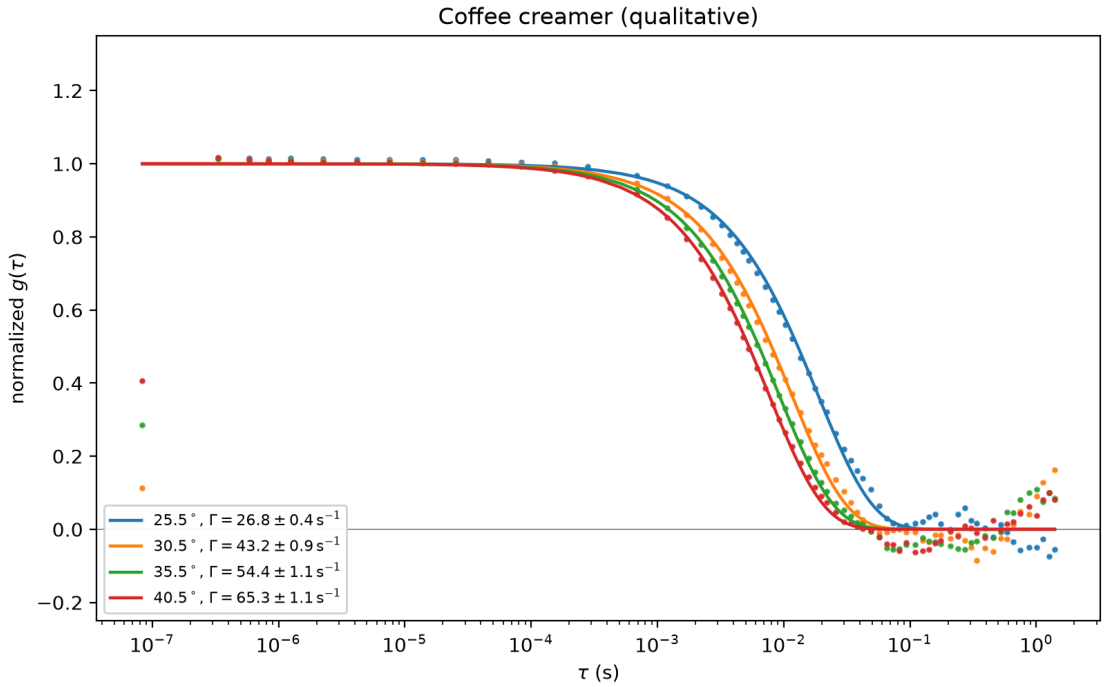


Figure 3.12: Multi-angle autocorrelation functions for diluted coffee creamer, with single-exponential fits weighted by analytic Poisson uncertainties $\sigma_g = \sqrt{g/N}$. Isolated short-lag points below the plateau are excluded from the fits; for this sample they appear below rather than above the plateau.

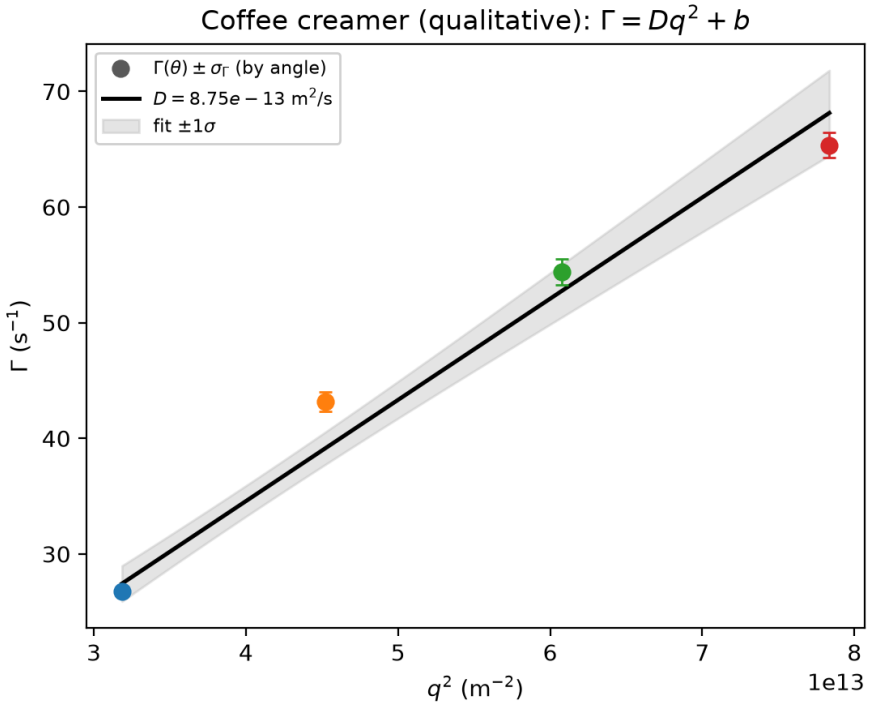


Figure 3.13: Decay rate Γ versus q^2 for diluted coffee creamer, with $\pm 1\sigma$ weighted-fit point uncertainties and a $\pm 1\sigma$ linear-fit band. Because the sample is polydisperse, the slope should be interpreted as an effective diffusion coefficient.

Together, the square-wave, latex-sphere, motor-noise, and coffee-creamer measurements demonstrate the performance of the FPGA-based DLS apparatus. The square-wave tests verify correlator timing. The latex-sphere measurements provide the primary quantitative validation through $\Gamma \propto q^2$. The motor-control tests identify and remove an instrumental artifact, while the coffee-creamer measurement shows operation on a complex nonideal sample.

CHAPTER 4

Discussion and Interpretation

This chapter interprets the measurements of Chapter 3: what the square-wave tests do and do not establish, how the latex diameters compare with nominal values, how the analytic uncertainty budget is constructed, and how the coffee-creamer result should be read.

4.1 Evaluation of test waveforms

The square-wave measurements provided a controlled timing test of the FPGA correlator. For input frequencies of 1 Hz, 100 Hz, and 10 kHz, the measured autocorrelation functions showed repeated peaks separated by the expected waveform periods, verifying the correlation algorithm over multiple timescales. These tests do not by themselves validate DLS performance: a square wave is deterministic and periodic, whereas photon arrivals in a DLS experiment are stochastic. The latex-sphere measurements provide the physical validation. Exploratory periodic square-wave tests near 100 kHz, noted in Chapter 3, did not recover a clean peak ladder and mark a limit of that timing check rather than of DLS photon correlation.

4.2 Hydrodynamic diameters of latex calibration standards

The latex-sphere measurements provide the primary validation of the FPGA-based DLS apparatus. For particles undergoing Brownian diffusion, the decay rate Γ should increase linearly with q^2. The observed linear relationships for the 90 nm, 300 nm, and 490 nm latex standards indicate that the measured autocorrelation

functions reflect physical particle motion rather than only electronic or optical artifacts.

Each intensity autocorrelation function was fit to the standard monodisperse DLS model

$$g_2(\tau) = 1 + \beta e^{-2\tau\Gamma}, \tag{1.22}$$

implemented in practice as $g(\tau) = B + \beta e^{-2\Gamma\tau}$ with a free baseline B. Plotting Γ against q^2 gives the diffusion coefficient from the slope of a linear fit. Using Eqs. 1.1 and 1.24, the hydrodynamic diameter $d = 2R_H$ was obtained for each particle size.

Table 4.1 summarizes the results. The 90 nm standard was recovered as 94 ± 8 nm (4% difference from nominal). The 300 nm and 490 nm standards were recovered as 397 ± 28 nm and 556 ± 64 nm (32% and 14% difference). In all most cases the correct size ordering is preserved, and Γ versus q^2 remains linear with $R^2 \geq 0.94$. Absolute agreement is strongest for the smallest standard. The quoted uncertainties are the combined fit-and-instrumental budget of Section 4.3; they do not include model bias such as residual multiple scattering or angular calibration.

The manufacturer lists a size-uniformity coefficient of variation (C.V.) of $\leq 15\%$ for the 0.09 μm standard and $\leq 3\%$ for the larger two (Table 2.2). That C.V. is the width of the particle-size distribution, not an uncertainty on the mean diameter, so it does not replace the measured error bars. It does indicate that a few-percent difference from the nominal 90 nm label is well within the stated distribution width ($\pm 15\%$ corresponds to about ± 14 nm).

Nominal (μm)	Calculated (μm)	Uncertainty (μm)	Difference (%)
0.09	0.094	0.008	4
0.30	0.397	0.028	32
0.49	0.556	0.064	14

Table 4.1: Hydrodynamic diameters from weighted single-exponential multi-angle analysis via $d = 2R_H$. Uncertainties are the combined analytic budget of Section 4.3.

4.3 Uncertainty budget

The uncertainty budget asks how much the reported diameters can change when counting noise and instrumental tolerances are included. As in Chapter 3, each lag stores coincidence counts C, and the interface computer forms $g = C/N$, where N is the total photon count for that acquisition. With C treated as Poisson, $\sigma_C = \sqrt{C}$. Because N is fixed for the run,

$$\sigma_g = \frac{\sigma_C}{N} = \frac{\sqrt{C}}{N} = \sqrt{\frac{g}{N}}.$$

Although N is the same at every lag, C (and therefore σ_g) still varies with τ. The single-exponential fit is a weighted least-squares fit with weights $1/\sigma_g^2$, so points with larger σ_g contribute less to the determination of Γ than points with smaller σ_g. The Poisson σ_g values are often smaller than the actual point-to-point scatter about the fitted curve. Following the usual practice when formal uncertainties underestimate the residuals, the uncertainties on Γ (and later on the Γ versus q^2 slope) are multiplied by a factor based on the reduced χ^2 so that they reflect the observed scatter, not only the Poisson estimate.

Instrumental contributions are estimated by shifting one prior at a time by its 1σ value, holding the fitted Γ values fixed, and recomputing D and d. The combined uncertainty is the root-sum-square of the statistical term and all instrumental contributions.

The 1σ priors used for the instrumental terms were: detector-angle offset $\pm 0.08^\circ$ (Section 1.6), wavelength ± 2 nm (spectrometer uncertainty at the operating drive voltage where the peak is 653 nm), refractive index $\pm 5 \times 10^{-4}$, temperature ± 1 K with water viscosity rescaled consistently with temperature, and an additional 1% residual viscosity uncertainty. Sample concentration was controlled using the dilution screen of Section 2.2 and Table 2.1, but concentration was not varied as a quantitative series in the multi-angle calibration runs and is not assigned a separate term here.

Table 4.2 lists the resulting contributions for the 90 nm run; the fractional breakdown is similar for the 300 nm and 490 nm runs. The Γ–q^2 slope uncertainty dominates. Temperature/viscosity is the leading instrumental term under these priors. The quoted $\pm 0.08°$ angle offset contributes only at the few-tenths-of-a-percent level over the measured angular range.

Source	Contribution (nm)	Relative (%)
Γ vs q^2 statistical	7.9	8.4
Temperature ($+\eta(T)$), ± 1 K	2.5	2.7
Viscosity residual, $\pm 1\%$	1.0	1.0
Wavelength, ± 2 nm	0.6	0.6
Angle offset, $\pm 0.08°$	0.2	0.2
Refractive index, $\pm 5 \times 10^{-4}$	0.1	0.1
Combined (90 nm)	8.3	8.9
Combined (300 nm)	28	7.0
Combined (490 nm)	64	12

Table 4.2: Uncertainty budget for the hydrodynamic diameter. Component rows are for the 90 nm latex run; combined rows give the final 1σ uncertainties for all three standards. Model bias (multiple scattering, non-exponential decay, and absolute angle error beyond the $\pm 0.08°$ prior) is not included.

The combined instrumental-and-fit uncertainties are about 7–12%. That budget covers the 4% offset of the 90 nm standard and is comparable to the 14% offset of the 490 nm standard, while the 32% offset of the 300 nm standard remains outside the quantified budget. Remaining candidates for that residual include multiple scattering, incomplete single-exponential decay, scattering-volume or alignment changes with angle, cuvette contamination, and a possible absolute angular calibration error larger than $\pm 0.08°$.

Figure 4.1 compares the measured diameters with the manufacturer nominal values. Vertical error bars are the combined analytic uncertainties from Table 4.2. Horizontal bands mark the manufacturer size-uniformity C.V. from Table 2.2 (distribution width, not an uncertainty on the mean). The 90 nm point agrees with nominal within the combined measurement uncertainty; the 490 nm point is consistent at about the 1σ level, while the 300 nm point remains high relative to its error bar.

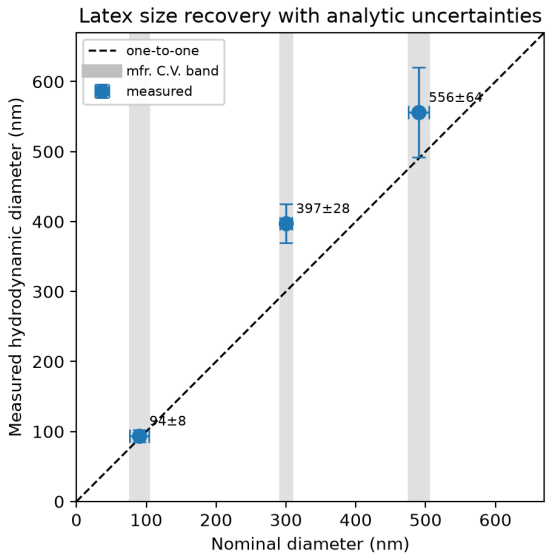


Figure 4.1: Measured hydrodynamic diameters versus nominal latex diameters. Vertical error bars are the combined analytic uncertainties of Section 4.3. Gray bands mark the manufacturer size-uniformity C.V. (Table 2.2). The dashed line is one-to-one agreement.

4.4 Coffee creamer as a qualitative test

The coffee-creamer measurement was a qualitative demonstration on a nonideal, polydisperse sample. A single-exponential fit therefore cannot be interpreted as a unique particle diameter; the effective size associated with the dominant relaxation is approximately 560 nm. Sample preparation was not optimized for DLS, and no attempt was made to fractionate the size distribution. The result shows only that the apparatus can acquire and fit multi-angle correlation data from a complex sample.

4.5 Overall interpretation

Taken together, these results support the central conclusion of the thesis: the FPGA-based autocorrelator can perform real-time correlation measurements suitable for multi-angle DLS. The square-wave tests verify the timing and correlation logic, while the latex-sphere measurements demonstrate recovery of diffusion coefficients from the expected Γ versus q^2 scaling under a weighted single-exponential analysis. The uncertainty budget of Section 4.3 accounts for the 90 nm offset and is comparable to the 490 nm residual, while the larger 300 nm residual points to unmodeled physical or calibration bias. The present apparatus should be viewed as a proof-of-concept system rather than a fully optimized commercial DLS instrument. Natural next steps are improved absolute angular calibration, direct temperature and viscosity metrology, and systematic tests of polydisperse samples with methods beyond a single exponential. Chapter 5 answers the motivating question directly.

CHAPTER 5

Conclusion

5.1 Can a multi-angle FPGA-based correlation device be effective for DLS?

Through experimental testing, this thesis has demonstrated that a real-time FPGA-based correlator can produce quantitative multi-angle dynamic light scattering (DLS) measurements at the proof-of-concept level. Initial validation using periodic test waveforms confirmed the correct timing and operation of the correlation algorithm. Measurements of calibrated latex-sphere suspensions showed the expected linear relationship between the decay rate Γ, obtained from weighted single-exponential fits, and the scattering vector q^2 (e.g. Fig 1.5). The resulting diffusion coefficients yielded hydrodynamic diameters of 94 ± 8, 397 ± 28, and 556 ± 64 nm for nominal 90, 300, and 490 nm standards, preserving the correct size ordering. The goal is not simply to obtain autocorrelation functions with the expected qualitative shape, but to use those functions to quantitatively measure Brownian diffusion. An analytic count-based and instrumental uncertainty budget at the 7–12% level accounts for the 90 nm offset and is comparable to the 490 nm residual, while the larger 300 nm residual points to calibration bias or an addition of other influences such as gravity causing particle movement to not be solely dependent on Brownian motion.

Although the apparatus successfully demonstrated quantitative DLS measurements, several improvements are required before it can be considered a fully validated research instrument. Improved absolute detector-angle calibration would reduce possi-

ble bias in the scattering vector. Automated motor-disable acquisition and improved electrical shielding would further reduce instrumental noise. More rigorous sample preparation and cuvette-cleaning procedures would reduce optical artifacts. A direct temperature and viscosity measurement would tighten the Stokes–Einstein conversion (Eq. 1.24), these measurements were based on the temperature of the room. Characterizing and mitigating detector afterpulsing—so that sub-microsecond lags can be used rather than excluded—is another natural direction for future work, particularly if smaller particles or larger scattering vectors push the physical correlation time closer to that artifact.

The FPGA architecture performs the correlation calculation in real time rather than requiring storage and offline processing of photon arrival data. This capability provides a flexible foundation for future extensions including higher count-rate operation, automated multi-angle measurements, cross-correlation DLS, modulated DLS, and eventually time-resolved or feedback-controlled scattering experiments.

In addition to validating the FPGA correlator itself, this work established a complete experimental methodology for multi-angle DLS, including custom 3D-printed optomechanical mounts for centering and driving the diffractometer, detector positioning, concentration optimization, calibration using latex standards, identification of instrumental artifacts, and quantitative extraction of diffusion coefficients with an analytic count-based and instrumental uncertainty budget. The present work does not establish the ultimate accuracy or precision limits of the instrument. Instead, it establishes that the measurement approach is physically valid and provides a foundation for future refinement.

The answer to the question posed at the beginning of this thesis is yes. Within the accuracy demonstrated here, the FPGA-based correlator successfully performed quantitative multi-angle DLS measurements and recovered physically meaningful diffusion coefficients from calibrated particle suspensions that showed a dependence on $\Gamma = Dq^2$.

APPENDIX A

Custom 3D-printed optomechanical components

Multi-angle DLS on this apparatus depends on several custom FDM-printed parts that commercial mounts do not provide. This appendix catalogs the seven designs highlighted in Section 1.7, with a photograph of each printed part and, where an STL model is available, a corresponding CAD view. The designs were iterated over the construction of the bench; the figures show representative as-printed hardware rather than every intermediate revision.

A.1 Limit-switch mount

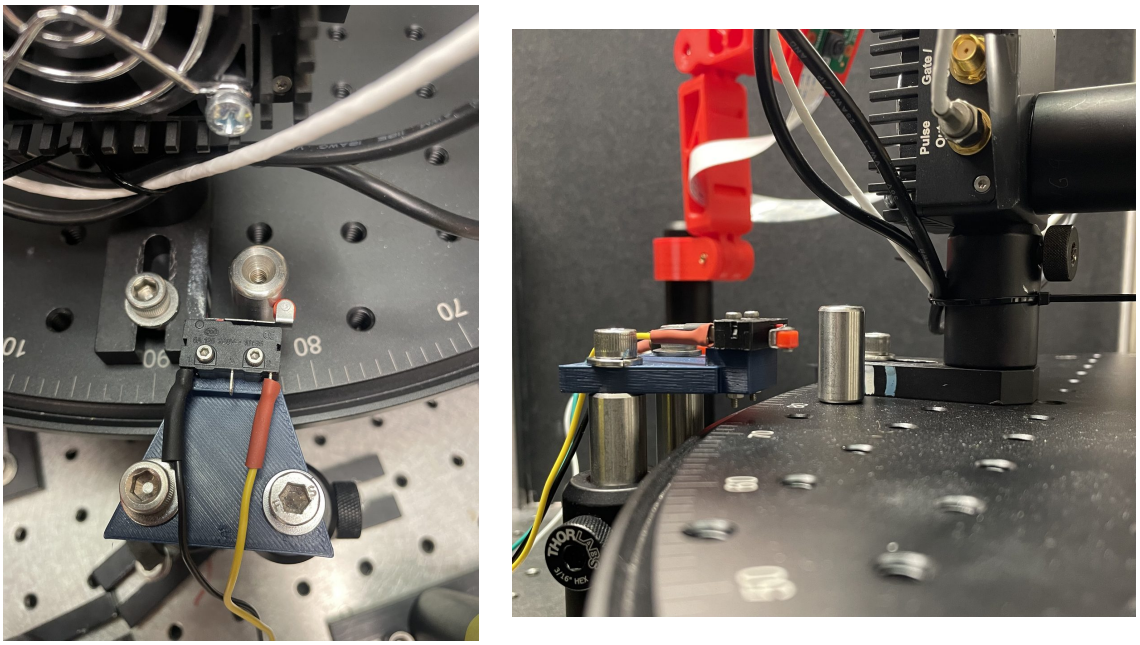


Figure A.1: Printed limit-switch mount bolted to the optical table (left) and engaged by the diffractometer reference post (right). Dual optical-table bolts stiffen the switch against bidirectional contact during homing.

Belt-driven rotation requires a mechanical origin after direction changes because the belt introduces backlash. The printed mount holds a limit switch on the optical-table grid so that a small post on the rotating assembly can trip the switch repeatably. Two table bolts provide lateral stiffness against the contact force in either travel direction. In practice two switches were used: one near the transmitted-beam direction as the angular origin, and a second farther along the arc (typically 100°–120° from the incident beam) as a software travel limit.

A.2 Diffractometer centering cap

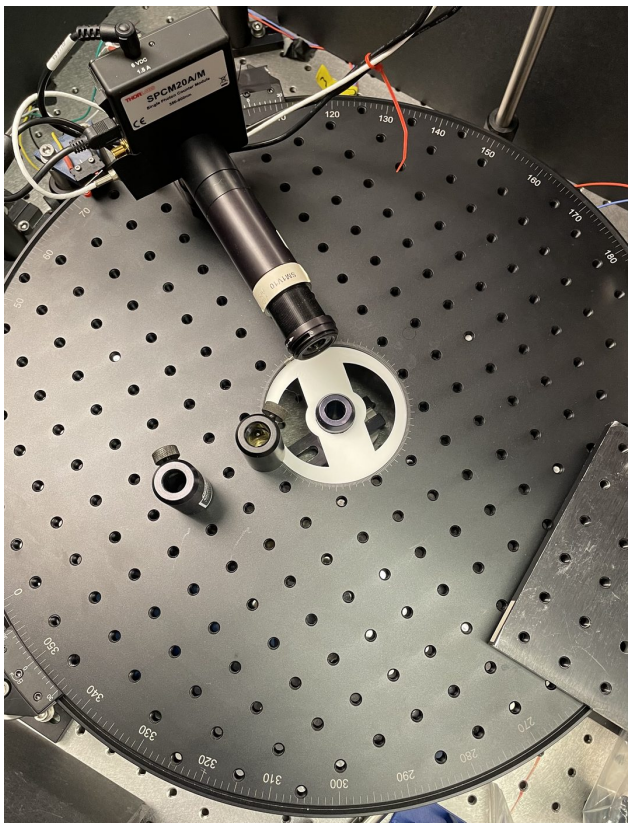
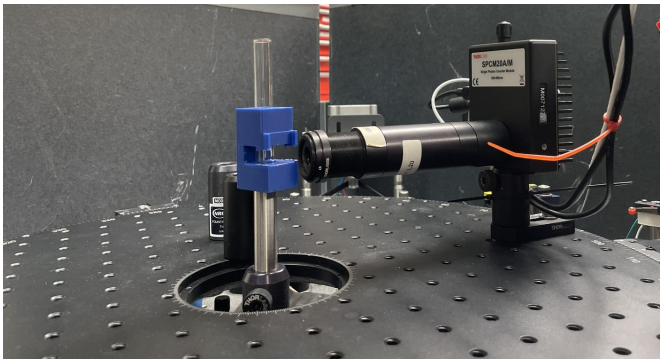


Figure A.2: Printed centering cap seated in the Thorlabs rotating-breadboard aperture. The outer diameter matches the removable center insert; the central sleeve hole locates the cuvette post on the rotation axis.

The Thorlabs large-area rotating breadboard has a removable central insert. The printed centering cap fits that aperture and provides a central hole for the post that carries the cuvette stage. Matching the outer diameter of the insert transfers the geometric center of the stage to the sample line, so that rotating the detector does not walk the observed scattering volume.

A.3 Cuvette stage



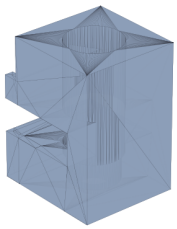


Figure A.3: Printed cylindrical cuvette stage with clearance windows for the incident, transmitted, and side-scattered beams (left). STL rendering of the cylindrical (12 mm) mount geometry (right).

Multi-angle collection favors a cylindrical cuvette so that the side wall presents a similar optical path as the detector sweeps in angle. The printed stage accepts a 12 mm-diameter Type 76 cuvette, centers it on the mounting bolt shared with the diffractometer axis, and leaves open windows for the laser path and for scattered light toward the detector. A square-cuvette variant and optional side windows (to restrict stray light in fixed-angle work) were also designed; the cylindrical open-window version was used for the multi-angle latex measurements.

A.4 Cuvette alignment pin

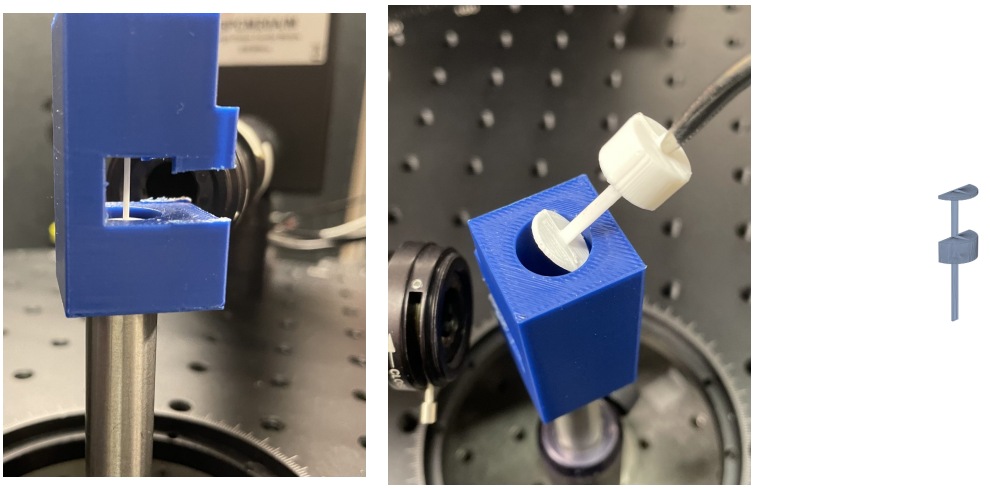


Figure A.4: Printed alignment pin: cylinder diameter matches the cuvette, with a thin rectangular scatterer on axis (left, center). STL rendering of one printable half (right); two halves are joined to form the finished pin.

Once the cuvette post is centered on the diffractometer, the laser must still be aimed through the same volume. The alignment pin occupies the cuvette seat and presents a thin rectangular prism on axis. Illuminating that edge with the diode laser provides a visible reference that couples the rotation center, the cuvette axis, and the beam path before liquid samples are loaded. Two printed halves are joined along the length to give a seamless outer cylinder.

A.5 Cuvette organizer

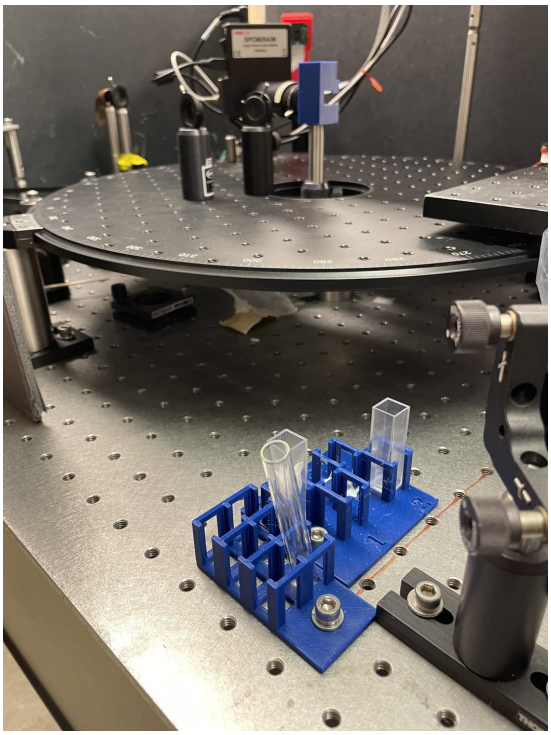


Figure A.5: Printed mountable cuvette organizer with numbered stations for sample preparation beside the optical table.

Sample preparation for DLS benefits from a stable, labeled place to rest cuvettes while dilutions are mixed and inspected. The printed organizer mounts to the optical table or sits free-standing and holds multiple cuvettes in numbered slots, reducing handling accidents and mix-ups during concentration screens.

A.6 Laser-to-optic mount

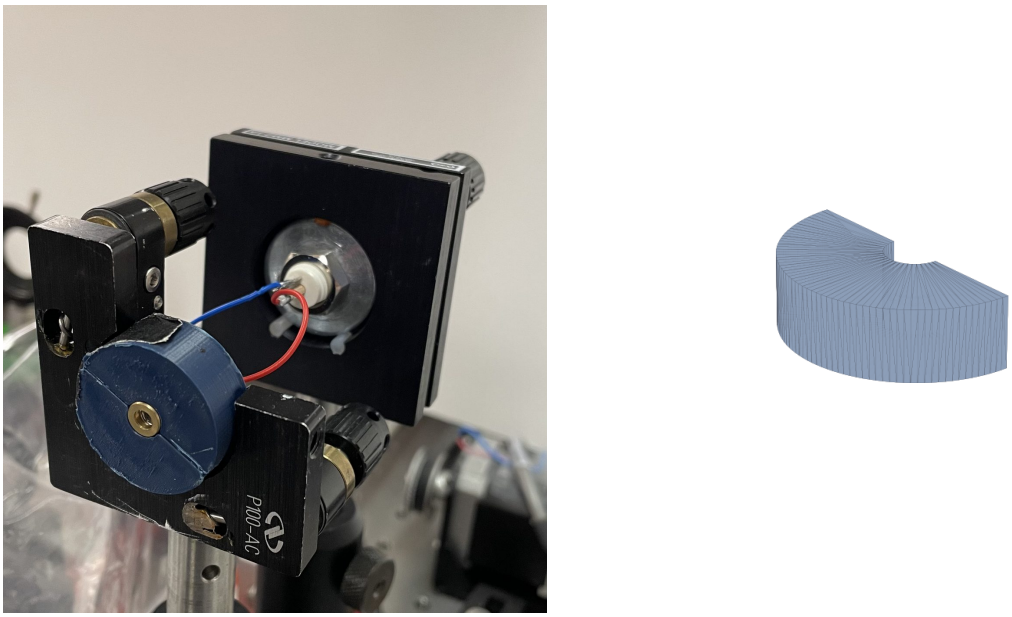


Figure A.6: Two-piece printed adapter holding the diode laser in a standard one-inch optic mount (left). STL rendering of one half, showing the internal ridge that seats the laser body (right).

The diode laser used for scattering is not a standard one-inch optic. The printed adapter splits into two halves that clamp the laser barrel; an internal ridge keys to the laser body. After assembly (with optional adhesive), the outer cylinder fits a conventional one-inch optic mount, so tip/tilt knobs can center the beam on the cuvette without a custom kinematic stage for the laser itself.

A.7 GT2 diffractometer gear



Figure A.7: Assembled printed GT2 gear that bolts under the rotating breadboard (left). STL rendering of one of the six printable arc segments, showing the GT2 tooth profile (right).

Angular fidelity of θ enters q and therefore the recovered diffusion coefficient. A commercial GT2 timing belt (repurposed from 3D-printer stock) couples a stepper motor to a printed ring gear bolted beneath the Thorlabs stage through the existing hole pattern. The gear was fabricated as six dovetailed segments that assemble into a full circle; adhesive was used during assembly for rigidity. The tooth profile suppresses belt slip that would otherwise appear as a non-reproducible angular offset between commanded and actual detector positions.

APPENDIX B

List of symbols and variables

The variables below appear in approximately the order they were introduced in the text.

Symbol	Variable
$I(t)$	Intensity signal
τ	Lag (delay) time
$G_2(\tau)$	Intensity autocorrelation (unnormalized)
$g_2(\tau)$	Normalized intensity autocorrelation
$g(\tau) = C(\tau)/N$	Correlator output (coincidences over acquisition total)
$C(\tau)$	Coincidence counts in lag channel τ
N	Total photon counts recorded during the acquisition
σ_g	Per-lag Poisson uncertainty on g, $\sqrt{C}/N = \sqrt{g/N}$
θ	Scattering angle
q, $\mathbf{q}$	Scattering-vector magnitude and vector
$g_1(\tau)$	First-order (field) autocorrelation
β	Optical coherence (contrast) factor
B	Fitted intensity-autocorrelation baseline
Γ	Field-correlation decay rate
b	Intercept in the Γ versus q^2 fit
D	Diffusion coefficient
n	Refractive index of the solvent
η	Sample viscosity
k_B	Boltzmann's constant
T	Temperature
R_H	Hydrodynamic radius
d	Hydrodynamic diameter ($d = 2R_H$)
C.V.	Coefficient of variation (relative width of a distribution)

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